MEDIPREP: HOSPITAL PROCEDURE PREPARATION AND READINESS SYSTEM

A Project Report submitted in partial fulfillment of the requirements for the award of the degree of

MASTER OF COMPUTER APPLICATIONS

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This is to certify that the project report entitled "MEDIPREP: HOSPITAL PROCEDURE PREPARATION AND READINESS SYSTEM" is a bonafide work carried out by [STUDENT NAME] (Register No.: [REGISTRATION NUMBER]) in partial fulfillment of the requirements for the award of the degree of MASTER OF COMPUTER APPLICATIONS by [AFFILIATED UNIVERSITY / INSTITUTION] during the academic year 2024 – 2026.

The project work embodies the original work of the candidate and has not been submitted previously for the award of any other degree, diploma, fellowship, or other similar title to this or any other University or Institution.

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I hereby declare that the project entitled "MEDIPREP: HOSPITAL PROCEDURE PREPARATION AND READINESS SYSTEM" submitted by me to the Department of Computer Applications, [COLLEGE / INSTITUTION NAME], in partial fulfillment of the requirements for the award of the Degree of Master of Computer Applications, is a record of bonafide project work carried out by me under the supervision and guidance of [GUIDE NAME], [Guide Designation], Department of Computer Applications.

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ABSTRACT

In modern healthcare administration, hospital surgical suites and diagnostic procedure rooms represent the most resource-intensive and financially critical assets. However, healthcare institutions across the globe routinely encounter substantial rates of avoidable, same-day procedure cancellations and delays—frequently exceeding 15% to 20% of scheduled elective interventions. The predominant root causes of these cancellations are administrative and perioperative communication failures, including incomplete pre-procedure preparations, missing or unverified diagnostic laboratory reports, unrecorded patient contraindications, and unconfirmed informed consents. Traditional manual preparation relies heavily on fragmented physical binders, disparate telephone follow-ups, and unverified patient assertions, leading to substantial clinical risk, compromised patient safety, and severe operational revenue loss.

To decisively resolve these operational vulnerabilities, this project introduces "MediPrep – Hospital Procedure Preparation and Readiness System", an enterprise-grade, web-based clinical workflow platform architected using the Python Flask web framework, MySQL relational database management system, and a robust 3-Tier Model-View-Controller (MVC) architectural design pattern. MediPrep establishes a tightly coordinated digital perioperative ecosystem encompassing four strictly governed user roles: System Administrators, Attending Physicians/Doctors, Scheduled Patients, and Hospital Laboratory Technicians. External diagnostic laboratories are recognized as third-party data origins but are strictly prohibited from possessing direct system authentication credentials, ensuring zero external exposure of internal hospital assets.

The core innovation of MediPrep lies in its dual-engine clinical governance framework: (1) a Dynamic Pre-Procedure Checklist Engine that automatically generates procedure-specific prerequisite timelines upon doctor scheduling, strictly requiring authorized Hospital Laboratory Technicians to upload official PDF reports with SHA-256 integrity hashes and doctor verification; and (2) a Deterministic Clinical Safety Rules Engine that evaluates vital metrics (blood glucose, hemoglobin, blood pressure, temperature, and creatinine) against clinical safety thresholds without relying on opaque or non-deterministic machine learning algorithms. The system synthesizes checklist task fulfillment, official report verification status, patient pre-op assessment submission, and signed informed consent into an automated, real-time Procedure Readiness Evaluation Engine. The readiness state is instantly communicated across doctor, patient, and clinical staff consoles via an intuitive color-coded paradigm (Green: Ready; Yellow: Pending Doctor Review; Red: Critical Incomplete / Contraindicated).

Complete system testing—incorporating unit testing, integration verification, acceptance scenarios, boundary value validation, and black-box/white-box testing across 16 formal test

cases—demonstrates 100% test success, deterministic rule execution, robust input sanitization, and full audit trail compliance. MediPrep provides healthcare facilities with an auditable, secure, and cost-effective digital solution that significantly minimizes perioperative delays, optimizes operating room utilization, and safeguards patient clinical outcomes.

Keywords: *Perioperative Readiness, Procedure Preparation, Clinical Safety Rules Engine, Hospital Information System, Role-Based Access Control, Flask MVC, Verification Pipeline, Audit Logging.*

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CHAPTER 1

INTRODUCTION

1.1 ABOUT THE PROJECT

In modern healthcare infrastructure, surgical suites and specialized interventional procedure rooms represent the most technically advanced, resource-intensive, and financially critical departments of any hospital or healthcare network. The seamless execution of scheduled elective and semi-urgent diagnostic and therapeutic procedures—such as Left Heart Catheterization, Upper Gastrointestinal Endoscopy, Colonoscopy, Laparoscopic Cholecystectomy, and Coronary Angiography—requires strict adherence to rigorous clinical preparation protocols before the patient ever enters the operative theater. These pre-operative requirements encompass multifaceted medical, biochemical, diagnostic, and legal prerequisites, including mandatory pre-anesthetic physiological assessments, specialized fasting timelines (Nil Per Os - NPO), cardiac evaluations (ECGs), hematological workups (Complete Blood Count, Coagulation Profiles), biochemical analyses (Fasting Blood Sugar, Serum Creatinine), and formally witnessed informed consent documentation.

Despite monumental advancements in clinical technology and surgical instrumentation, the perioperative management workflow in a vast majority of healthcare institutions remains shockingly antiquated. Traditional workflows depend almost entirely on physical paper binders, handwritten clinical slips, uncoordinated telephone inquiries, and disjointed departmental communications. Consequently, healthcare facilities globally experience an alarming rate of avoidable, same-day surgical cancellations and costly operating room delays—frequently ranging between 12% to 22% of scheduled elective surgical cases according to contemporary clinical studies. The root causes of these cancellations are overwhelmingly administrative and perioperative communication breakdowns rather than acute surgical emergencies: misplaced diagnostic lab reports, unverified external laboratory slips, unrecorded patient allergies, lack of documented informed consent, or sudden discovery of severe clinical contraindications (such as acute hyperglycemia or hypertensive crisis) only after the patient has arrived on the operating table.

The financial and human consequences of such cancellations are severe. For healthcare institutions, an idle operating room represents an average financial loss of $1,000 to $2,500 per hour in unrecoverable overhead, wasted staffing time, and surgical slot forfeiture. For patients and their families, last-minute procedural cancellations lead to extreme emotional distress, prolonged physical suffering, unnecessary pre-procedural fasting, missed wages, and escalated health risks associated with deferred medical interventions.

To decisively overcome these acute operational vulnerabilities, this project introduces "MediPrep – Hospital Procedure Preparation and Readiness System". MediPrep is an enterprise-grade, web-based clinical workflow automation and procedural readiness platform developed utilizing the Python Flask web framework, MySQL relational database management system, and a strictly disciplined 3-Tier Model-View-Controller (MVC) software architectural pattern. MediPrep centralizes the entire perioperative journey into a unified, role-governed digital ecosystem encompassing Hospital Administrators, Attending

Doctors/Physicians, Scheduled Patients, and Hospital Laboratory Technicians. By digitizing dynamic checklist management, enforcing an uncompromised lab verification pipeline, evaluating vital thresholds via a deterministic clinical safety rules engine, and calculating real-time visual readiness status (Green, Yellow, Red), MediPrep bridges the critical communication chasm between patients, diagnostic laboratories, and surgical teams.

> **Key Philosophy of MediPrep**
> MediPrep enforces deterministic, auditable clinical safety rules. The system deliberately rejects unpredictable 'black-box' AI algorithms in favor of transparent, clinically validated physiological boundary checks and immutable verification workflows that provide surgical teams with 100% confidence on procedure day.

1.2 OBJECTIVES OF THE PROJECT

The primary objective of MediPrep is to engineer and deploy an integrated, secure, and user-centric perioperative readiness platform that eliminates preventable same-day procedural cancellations while enhancing patient safety and hospital operational efficiency. The specific architectural and clinical objectives include:

- **1. Eradication of Preventable Procedure Cancellations:** To provide a centralized digital platform that systematically tracks every pre-operative prerequisite—from clinical assessments and fasting schedules to diagnostic tests and consent forms—ensuring zero unexpected omissions on the scheduled procedure date.
- **2. Role-Based Perioperative Workflow Orchestration:** To establish strictly governed, role-specific operational dashboards for Administrators, Doctors, Patients, and Hospital Laboratory Technicians, ensuring each stakeholder has targeted visibility and operational capabilities appropriate to their clinical authority.
- **3. Enforcing an Official Laboratory Verification Pipeline:** To implement a strict clinical security protocol wherein patients are strictly prohibited from uploading unverified lab results directly. All official diagnostic reports (CBC, Blood Sugar, ECG, Renal Function) must be uploaded exclusively by authorized Hospital Laboratory Technicians—with SHA-256 file hashes, provenance tracking, and formal Doctor verification—guaranteeing data authenticity.
- **4. Deterministic Clinical Safety Rules Evaluation:** To engineer an automated, rule-based clinical safety engine that evaluates vital physiological metrics (Blood Sugar >200 mg/dL, Hemoglobin <10 g/dL, Systolic BP >140 mmHg, Diastolic BP >90 mmHg, Body Temperature >100.4°F or <95.0°F, Creatinine >1.5 mg/dL) and automatically generates prioritized risk alerts for attending clinical teams.
- **5. Real-Time Procedure Readiness Scoring & Visualization:** To develop an intelligent readiness evaluation algorithm that synthesizes the real-time status of checklist items, lab verification, assessment locking, and clinical vitals into an instantaneous, color-coded visual

indicator (Green: Ready for Procedure; Yellow: Pending Doctor Review; Red: Incomplete / Critical Contraindication).

- **6. Transparent Multi-Channel Communication and Clarification:** To establish a closed-loop digital clarification channel allowing attending physicians to communicate directly with laboratory technicians regarding ambiguous or abnormal test results, resolving clinical ambiguities prior to patient admission.
- **7. Comprehensive Audit Logging and Legal Accountability:** To maintain an immutable, tamper-evident audit log of every system transaction—including user authentications, procedure schedulings, report uploads, doctor verifications, vitals modifications, and status overrides—ensuring full compliance with healthcare regulatory standards.

1.3 SCOPE OF THE PROJECT

The scope of MediPrep encompasses the comprehensive digitalization and procedural governance of hospital pre-operative preparation across secondary and tertiary healthcare facilities, surgical specialty clinics, and day-care ambulatory surgery centers. The boundaries and operational dimensions of the system are defined across several structural tiers:

1.3.1 In-Scope Capabilities

- **Multi-Specialty Procedure Catalog:** Support for an extensible procedural repository accommodating cardiovascular, gastrointestinal, orthopedic, and general surgical interventions, each mapped to customizable dynamic pre-procedure checklist templates and room allocations.
- **Patient Pre-Operative Engagement:** Empowering patients to register securely, review customized procedure schedules, access personalized preparation guidelines, complete comprehensive medical history and anesthesia questionnaires, and digitally confirm informed procedural consents.
- **Clinical and Laboratory Coordination:** Providing dedicated consoles for attending physicians to schedule procedures, track patient compliance, review laboratory reports, record clinical vitals, resolve lab clarifications, and perform authoritative procedure clearance.
- **Enterprise Security and Relational Integrity:** Enforcing industry-standard password hashing (PBKDF2/SHA-256 via Werkzeug), secure session management, input boundary validation, foreign-key relational constraints across 20 normalized MySQL tables, and strict Role-Based Access Control (RBAC).

1.3.2 Out-of-Scope Elements and Future Boundaries

To maintain architectural focus, high reliability, and rapid deployment within the academic project lifecycle, certain advanced tertiary integrations are designated as out-of-scope for the current release: direct real-time bidirectional integration with legacy proprietary Electronic Health Record (EHR) systems via HL7/FHIR protocols, third-party SMS/WhatsApp telecommunication gateways (simulated in-app notifications are implemented), automated

intraoperative anesthesia monitoring, and post-discharge recovery tracking. These capabilities represent natural extensions for subsequent enterprise phases as outlined in Chapter 10.

CHAPTER 2

INITIAL INVESTIGATION AND FEASIBILITY STUDY

2.1 INITIAL INVESTIGATION

The initial investigation phase was undertaken to analyze the existing perioperative preparation procedures currently practiced across public and private healthcare institutions. A comprehensive operational investigation involving interviews with surgical coordinators, nursing supervisors, hospital laboratory technicians, and practicing physicians revealed significant structural vulnerabilities in the traditional pre-operative preparation workflow:

- **Severe Information Fragmentation:** Patient information is routinely scattered across disjointed physical records, outpatient consultation slips, independent diagnostic center envelopes, and disconnected hospital software modules. As a result, attending surgical teams have no unified overview of a patient's overall readiness until hours before the scheduled operation.
- **Reliance on Patient Memory and Verbal Assertions:** Nurses and surgical staff frequently rely on verbal questioning to determine whether patients followed critical preparation instructions (such as 8-hour fasting or cessation of anticoagulant medications). Patients often misinterpret clinical terms, leading to dangerous situations where patients are sedated before staff realize fasting guidelines were violated.
- **Delayed Discovery of Abnormal Laboratory Values:** Routine diagnostic tests (e.g., blood sugar, hemoglobin, clotting times) conducted by external diagnostic centers or hospital laboratories are frequently presented to the surgical team only during the pre-operative holding area intake. When severe abnormalities (such as blood glucose exceeding 250 mg/dL or hemoglobin under 8 g/dL) are detected at this late stage, the surgery must be abruptly aborted, wasting operating theater time and surgical crew availability.
- **Lack of Provenance and Quality Assurance for Diagnostic Reports:** In many hospitals, patients physically bring reports from various external laboratories. These reports vary widely in format, lack standardized verification, and present authentication risks. The hospital lacks an established mechanism for doctors to seek clarifications from laboratory personnel prior to the day of surgery.
- **Absence of Objective Readiness Criteria:** There is no standardized, objective scoring metric to classify whether a patient is 'Green' (fully cleared), 'Yellow' (pending minor non-critical verifications), or 'Red' (critical contraindication or incomplete preparation). Decisions are made ad-hoc, increasing cognitive burden on surgeons and anesthesiologists.

The findings of this initial investigation unequivocally demonstrated the urgent imperative for a centralized, web-based digital preparation tracking platform that automates checklist workflows, coordinates laboratory uploads, evaluates clinical safety rules, and computes objective readiness scores in real time.

2.2 FEASIBILITY STUDY

Before proceeding with system architecture and software development, a rigorous feasibility study was conducted to ascertain the economic, technical, behavioral, and operational viability of the proposed MediPrep system. The feasibility study ensures that the software

can be practically deployed, maintained, and operated within the constraints of real-world healthcare settings.

2.2.1 Economic Feasibility

Economic feasibility assesses the financial justification of the project by performing a thorough cost-benefit analysis. The costs associated with developing and deploying MediPrep are divided into hardware infrastructure, software licensing, and operational maintenance. By strategically adopting open-source technologies—Python, Flask, MySQL, Bootstrap, and Apache—the software licensing cost is precisely zero.

Table 2.1: Cost Estimation & Economic Feasibility Analysis

COST COMPONENT	DESCRIPTION & SPECIFICATION	ESTIMATED COST (INR)
Server Hardware	Standard enterprise multi-core application/database server	₹ 65,000
Operating System	Ubuntu Linux Server LTS (Open Source)	₹ 0
Database Management	MySQL Community Edition / MariaDB (GPL License)	₹ 0
Programming & Framework	Python 3.10+ and Flask Micro-framework (BSD License)	₹ 0
Development Tools	Visual Studio Code, Git, Chrome DevTools	₹ 0
Networking & SSL	Standard hospital LAN infrastructure & Let's Encrypt SSL	₹ 5,000
Training & Deployment	Internal staff training and orientation sessions	₹ 15,000
Annual Maintenance	Routine database backup, security updates, and patching	₹ 20,000
TOTAL ESTIMATED COST	Comprehensive first-year system implementation cost	₹ 1,05,000

Conversely, the tangible financial benefits of MediPrep are substantial. Operating room idle time costs average ₹ 15,000 to ₹ 35,000 per hour in Indian tertiary hospitals. By preventing just 5 to 10 avoidable procedural cancellations per month, MediPrep delivers an estimated annual operational savings of ₹ 9,00,000 to ₹ 21,00,000. Therefore, the return on investment (ROI) is achieved within the first 2 to 3 months of hospital-wide adoption, establishing overwhelming economic feasibility.

2.2.2 Technical Feasibility

Technical feasibility evaluates whether the project can be constructed using available technology, developmental expertise, and computational infrastructure. MediPrep leverages a robust, battle-tested modern web stack:

- **Server-Side Architecture:** Python 3.10+ and the Flask micro-framework provide exceptional development agility, modular Blueprint routing, built-in WSGI support, and extensive libraries for security, cryptographic hashing, and data serialization.
- **Database Reliability:** MySQL 8.0 with the InnoDB storage engine guarantees strict ACID compliance, robust referential integrity across foreign key constraints, and high-concurrency transactional throughput required for medical record storage.
- **Client-Side Ubiquity:** HTML5, CSS3, JavaScript (ES6), and Bootstrap 5 ensure complete responsive compatibility across desktop workstations, clinical tablet computers, and mobile smartphones without requiring end-users to install proprietary native applications.
- **Hardware Accessibility:** The system operates seamlessly on standard commodity PC hardware (Intel Core i3/i5, 8 GB RAM) and requires no high-end GPU or specialized computational clusters, guaranteeing high technical feasibility.

2.2.3 Behavioral Feasibility

Behavioral feasibility examines whether end-users (administrative staff, physicians, laboratory technicians, and patients) will accept, adopt, and effectively interact with the new system. User resistance in healthcare typically stems from steep learning curves and excessive administrative data-entry burdens. MediPrep directly resolves behavioral friction through targeted design strategies:

- **Minimalist Clinical Interfaces:** Doctor and laboratory consoles present clean, uncluttered tables, auto-populating patient demographics and highlighting only pending tasks, thereby minimizing cognitive fatigue.
- **Patient-Centric Portal:** The patient interface features simple, jargon-free checklist items, clear step-by-step progress bars, and mobile-friendly input cards, enabling even elderly or non-tech-savvy individuals to complete pre-op forms with ease.
- **Role-Specific Operational Clarity:** Technicians are only presented with lab upload tasks; doctors only see scheduling and verification queues; administrators focus on user management and audit inspection. This segregation prevents confusion and promotes high user adoption.

2.2.4 Operational Feasibility

Operational feasibility assesses how comfortably the software aligns with the day-to-day organizational procedures and standard operating protocols (SOPs) of the hospital. MediPrep was explicitly modeled around standard hospital surgical preparation workflows:

- **Alignment with Clinical Governance:** The multi-step verification pipeline (Technician uploads official report -> Doctor verifies -> System evaluates readiness) mirrors the existing legal and clinical accountability chains of modern hospitals.
- **Audit Compliance:** Every single user interaction is permanently recorded in tbl_audit_logs, satisfying the regulatory requirements of hospital accreditation bodies (such as NABH in India and JCI internationally).
- **Staff Shift Handover Continuity:** Because all pre-op statuses, reports, and risk notes are stored centrally in real time, oncoming nursing and surgical shifts can instantly assess patient readiness without relying on verbal handovers. The system is therefore completely operationally feasible.

CHAPTER 3

SYSTEM ANALYSIS

3.1 SYSTEM ANALYSIS

System analysis is the comprehensive dissection of a system into its constituent components to study how well those parts work and interact with each other to accomplish their designated purpose. In this project, system analysis involved a detailed investigation of the perioperative journey from the initial surgical consultation to the moment of theater transfer, identifying vulnerabilities in current practices and establishing exact specifications for the proposed MediPrep platform.

3.2 EXISTING SYSTEM

The existing pre-operative preparation workflow in conventional healthcare institutions relies heavily on manual, paper-based, and fragmented communication mechanisms. When an elective procedure is scheduled, the patient is provided with a stack of physical paper slips containing pre-procedure instructions, fasting directives, laboratory requisition forms, and consent documents. The operational lifecycle of this existing system is fraught with inherent risks and procedural friction:

- **Physical Documentation Loss:** Patients frequently misplace laboratory requisition slips or leave completed diagnostic reports at home on the morning of surgery, forcing hospital staff to duplicate expensive tests or cancel the scheduled procedure.
- **Zero Real-Time Visibility:** Surgeons and operating room coordinators have no mechanism to track whether an outpatient has completed necessary diagnostic tests until the patient physically arrives at the hospital on the morning of the procedure.
- **Unverified Patient Uploads & Alteration Vulnerabilities:** In basic digitized setups where patients are allowed to upload their own lab reports, patients frequently upload irrelevant documents, unreadable blurry photos, or unverified documents, introducing catastrophic clinical and legal hazards.
- **Delayed Anesthetic & Contraindication Discovery:** Vital signs and diagnostic test results are checked only during pre-anesthetic intake on procedure day. Dangerous contraindications (severe anemia, acute fever, critical hyperglycemia, or high serum creatinine indicating renal impairment) are identified too late to manage, forcing emergency surgical rescheduling.
- **Disjointed Communication Loops:** When an attending doctor questions an ambiguous laboratory result, resolving the issue requires multiple telephone calls to the laboratory department. If the pathologist or technician is unavailable, the communication thread is broken without an audit trail.

3.3 PROPOSED SYSTEM

The proposed "MediPrep – Hospital Procedure Preparation and Readiness System" completely supplants the flawed manual methodology with an integrated, automated, and secure digital platform. MediPrep establishes a single source of perioperative truth accessible to all accredited stakeholders:

- **Centralized Procedure Catalog & Dynamic Checklists:** Attending physicians select from standardized procedure catalogs (e.g., Left Heart Catheterization, Upper GI Endoscopy). Upon scheduling, the system automatically instantiates a tailored pre-procedure checklist with precise due dates and clinical criticality flags.
- **Secured Official Laboratory Upload Pipeline:** Patients are strictly restricted from uploading unverified clinical laboratory reports. Only accredited Hospital Laboratory Technicians possess credentials to upload official PDF diagnostic reports, recording laboratory name, technician identity, and SHA-256 hash.
- **Mandatory Doctor Verification & Clarification Channel:** Every diagnostic report must be formally verified and approved by the attending physician. If a report is ambiguous, the doctor can open a real-time digital clarification request directly linked to the specific test item.
- **Automated Clinical Safety Rules Engine:** A deterministic rules engine evaluates patient vitals against clinical safety thresholds, automatically flagging hyperglycemia, severe anemia, hypertension, fever, and renal impairment.
- **Dynamic Procedure Readiness Scoring (Green/Yellow/Red):** The platform continuously computes overall procedure readiness. A patient is marked 'Green' (Ready) only when all checklist tasks are completed, reports verified, pre-op assessment locked, consent confirmed, and vitals are within safe bounds.
- **Comprehensive Audit Logging:** Every single action across the system is recorded in tbl_audit_logs with user identity, role, timestamp, action, and affected patient/procedure ID.

3.4 SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

The Software Requirements Specification (SRS) establishes a complete description of the behavior and constraints of the MediPrep platform. It details the functional, non-functional, interface, and architectural requirements of the system.

3.4.1 Scope of the Project

MediPrep governs pre-operative preparation workflows for hospital surgical and interventional departments. It encompasses four dedicated user roles (Administrator, Doctor, Patient, Hospital Lab Tech), enforces strict laboratory verification pipelines, executes automated physiological safety rules, and generates real-time procedural readiness scores.

3.4.2 Overall Description

MediPrep operates as a web-based client-server application hosted on a central hospital server and accessed via standard web browsers. All data is stored in a normalized MySQL relational database. The user interface is fully responsive, catering to desktop workstations in clinical offices, mobile tablets in pre-op wards, and smartphones used by outpatients at home.

3.4.3 Functional Requirements

- **SRS-FR-01: User Authentication & Role-Based Access:** The system shall authenticate users via unique credentials, enforce PBKDF2/SHA-256 password hashing, validate strict Indian mobile number formats (10 digits starting with 6-9), enforce DOB strictly in the past, and restrict module access based on verified role IDs.
- **SRS-FR-02: Procedure Catalog & Scheduling Management:** The system shall allow Doctors to schedule procedures for registered patients, assign preparation and procedure rooms, and define target execution dates and times.
- **SRS-FR-03: Dynamic Checklist Instantiation:** Upon procedure scheduling, the system shall automatically generate customized pre-procedure checklist items derived from master procedure templates, assigning task types (patient action, staff verification, document upload) and criticality flags.
- **SRS-FR-04: Official Laboratory Report Upload:** The system shall permit only authorized Hospital Laboratory Technicians to upload official PDF diagnostic reports, recording laboratory origin, test metadata, file size, and cryptographic hash.
- **SRS-FR-05: Doctor Report Verification & Clarification:** The system shall enable attending doctors to review uploaded reports, mark them as 'verified', or initiate a structured clarification request to the laboratory technician.
- **SRS-FR-06: Pre-Op Assessment & Digital Consent:** The system shall allow patients to complete comprehensive pre-procedure health questionnaires, lock submitted assessments, and review and digitally confirm legally binding procedure consents.
- **SRS-FR-07: Clinical Safety Rules Evaluation:** The system shall automatically evaluate recorded patient vitals against deterministic clinical thresholds (Blood Sugar >200 mg/dL, Hemoglobin <10 g/dL, BP >140/90 mmHg, Temp >100.4°F or <95.0°F, Creatinine >1.5 mg/dL) and trigger clinical warning alerts.
- **SRS-FR-08: Procedure Readiness Scoring & Audit Trail:** The system shall compute real-time readiness status (Green, Yellow, Red) based on checklist compliance and clinical safety rules, while logging every transaction in an immutable audit table.

3.4.4 Non-Functional Requirements

- **SRS-NFR-01: Security & Confidentiality:** Role-based access control decorators shall protect every administrative and clinical endpoint. Session cookies must be HTTP-only and protected against cross-site scripting (XSS) and SQL injection via parameterized prepared statements.
- **SRS-NFR-02: Performance & Response Latency:** The system shall maintain an average HTTP response latency of under 500 milliseconds for standard queries and complete readiness re-calculations within 100 milliseconds upon database updates.
- **SRS-NFR-03: Reliability & Data Integrity:** All database operations shall execute within ACID-compliant transactional boundaries, preventing partial checklist creation or orphaned report records.

- **SRS-NFR-04: Usability & Accessibility:** The interface shall adhere to WCAG 2.1 accessibility standards, utilizing high-contrast typography, clear iconography, and responsive layouts adaptable across mobile, tablet, and desktop viewports.
- **SRS-NFR-05: Maintainability & Modularity:** The codebase shall adhere to strict MVC layering, segregating routing controllers, domain business logic services, and data access repositories to ensure independent maintainability.
- **SRS-NFR-06: Auditability:** The system shall guarantee non-repudiation by logging all user actions with exact actor IDs, role names, timestamps, and target entities.

3.4.5 User Interface Requirements

The user interface must be clean, responsive, and intuitive. It utilizes a professional healthcare color palette: Navy Blue (#1B365D) for navigation and primary branding, Slate Blue (#2C5282) for secondary elements, and unambiguous clinical status badges: Emerald Green (#10B981) for 'Ready', Amber Yellow (#F59E0B) for 'Pending Review', and Crimson Red (#EF4444) for 'Not Ready / Contraindicated'.

3.4.6 System Architecture

MediPrep implements a classic 3-Tier Model-View-Controller (MVC) architecture:
1. Presentation Tier (Views): Composed of responsive HTML5 templates rendered via Jinja2, Bootstrap 5 UI grids, and client-side JavaScript.
2. Application / Logic Tier (Controllers & Services): Implemented in Python Flask, managing route handling, session authentication, the Clinical Safety Rules Engine, and the Readiness Evaluation Service.
3. Data Tier (Models & MySQL): A normalized relational schema of 20 tables managed by MySQL InnoDB, enforcing relational constraints, cascading updates, and transaction rollbacks.

3.4.7 Conclusion

The system analysis confirms that MediPrep provides a robust, scalable, and technically sound solution to the critical perioperative coordination challenges faced by modern healthcare institutions. By establishing clear functional boundaries, deterministic safety rules, and an uncompromised verification pipeline, the platform is optimally engineered to ensure surgical readiness and patient safety.

CHAPTER 4

MODULES

4.1 MODULES DESCRIPTION

The MediPrep system is architected into seven distinct, highly cohesive, and loosely coupled functional modules. Each module addresses a specialized domain of the perioperative workflow while seamlessly interacting with the central relational data store and the clinical rules engine. The following subsections describe the technical architecture, business logic, and operational responsibilities of each module.

4.1.1 User Management & Role-Based Access Control (RBAC)

The User Management and RBAC module governs authentication, credential verification, identity provisioning, and access authorization across the platform. MediPrep defines four strictly separated user roles stored in tbl_roles:
1. Administrator (Role ID 1): Exercises global management privileges over user accounts, doctor profiles, procedure master catalogs, checklist templates, and the complete institutional audit trail.
2. Doctor / Attending Physician (Role ID 2): Schedules procedures, monitors patient compliance, reviews and formally verifies diagnostic laboratory reports, records clinical vitals, initiates lab clarifications, and performs procedure clearance.
3. Patient (Role ID 3): Accesses personalized procedure schedules, reviews fasting and medication instructions, completes pre-procedure health questionnaires, and confirms digital informed consents.
4. Hospital Laboratory Technician (Role ID 4): Inspects pending investigation queues and uploads official PDF laboratory reports with cryptographic hashes. External laboratories do not possess user accounts; their reports are ingested exclusively through authorized technicians.

- **Strict Input Boundary Validation:** During registration, the module enforces stringent clinical validation rules: (a) Date of Birth must be strictly in the past, rejecting today's date or future dates; (b) Mobile numbers must adhere strictly to the 10-digit Indian telecommunications format starting with 6, 7, 8, or 9; (c) Password security enforces a minimum of 8 characters without spaces; (d) Duplicate user validation checks ensure unique usernames and email addresses across the database without wiping out valid form fields on failure.
- **Cryptographic Security & Route Protection:** User passwords are encrypted using PBKDF2 with SHA-256 password hashing via Werkzeug security primitives. All protected routes are guarded by custom Python decorators: @login_required verifies active session state, while @role_required(allowed_roles) ensures users cannot escalate privileges or access unauthorized clinical consoles.

4.1.2 Procedure Catalog & Scheduling Management

This module manages the master repository of medical and surgical interventions (tbl_procedures) and coordinates procedural scheduling (tbl_scheduled_procedures). Each

procedure entry contains a unique clinical code, procedural name (e.g., Left Heart Catheterization, Upper GI Endoscopy), category, detailed clinical description, and estimated operative duration.

- **Attending Physician Scheduling Workflow:** When an attending doctor schedules a procedure, they select an existing registered patient, specify the clinical procedure, designate the procedure date and target time, and allocate specific preparation rooms and operative theaters (tbl_procedure_rooms).
- **Relational Synchronization:** Upon submission, the system records the transaction in tbl_scheduled_procedures, establishing foreign key linkages with tbl_users, tbl_doctor_profiles, and tbl_procedures. This automatically triggers downstream events: initializing the dynamic checklist, creating empty vitals records, and provisioning the readiness evaluation record.

4.1.3 Dynamic Pre-Procedure Checklist & Task Management

To ensure comprehensive preparation, MediPrep implements a dynamic, template-driven pre-operative checklist engine. Rather than utilizing static paper forms, the system automatically instantiates a tailored checklist (tbl_checklist_items) derived from predefined procedure templates (tbl_dynamic_checklist_templates) at the exact moment a procedure is scheduled.

- **Categorized Checklist Task Types:** Checklist items are categorized into three operational types: (1) 'patient_action' for patient-performed tasks such as 8-hour NPO fasting, stopping blood-thinning medications, and surgical skin preparation; (2) 'staff_verification' for nursing and pre-op staff tasks such as verifying IV access, allergies, and baseline vitals; and (3) 'document_upload' for mandatory laboratory diagnostic reports.
- **Dynamic Timeline Calculation & Criticality Flags:** Each task template defines a due_days_before parameter. The system computes the exact deadline (due_by_datetime) relative to the scheduled procedure date. High-risk prerequisites are marked with an is_critical flag; incomplete critical items immediately block procedure clearance in the readiness engine.

4.1.4 Official Lab Reports & Laboratory Verification Workflow

A critical vulnerability in healthcare systems is the ingestion of inaccurate, altered, or unverified laboratory reports. MediPrep resolves this through an uncompromised, role-restricted laboratory upload and verification pipeline (tbl_uploaded_reports).

- **Strict Prohibition of Patient Lab Uploads:** Under the MediPrep security policy, patients are strictly prohibited from uploading official laboratory reports directly. This eliminates the danger of unreadable photos, altered values, or fake laboratory slips entering the clinical record.
- **Authorized Technician Report Ingestion:** Official diagnostic PDF reports (Complete Blood Count, Blood Glucose, ECG, Renal Function) must be uploaded exclusively by

authorized Hospital Laboratory Technicians. The technician specifies the issuing laboratory (e.g., Central Hospital Clinical Laboratory or accredited external partner ABC Diagnostics), external report ID, test date, and uploads the verified PDF document. The system calculates a SHA-256 cryptographic file hash to guarantee document immutability.

- **Two-Tier Doctor Verification & Clarification Loop:** Uploaded reports enter a 'pending_verification' status. The attending physician inspects the PDF report directly within their clinical console. The doctor can either approve the report ('verified') or open a real-time clarification ticket (tbl_lab_clarifications). The technician responds directly to the query, providing complete non-repudiation and transparency before the report is verified.

4.1.5 Patient Vitals, Clinical Assessments & Safety Rule Engine

Patient safety demands rigorous screening for physiological contraindications before anesthesia and surgical intervention. MediPrep combines patient-reported subjective assessments (tbl_patient_assessments) with objective clinical vital measurements (tbl_patient_vitals) and executes an automated Clinical Safety Rules Engine.

- **Pre-Procedure Clinical Assessment:** Patients complete a detailed pre-operative questionnaire detailing known drug allergies, active medications, prior surgical complications, anesthesia adverse reactions, and emergency contact details. Once submitted, the assessment is permanently locked ('submitted_locked') to prevent retrospective tampering.
- **Deterministic Clinical Safety Rules Engine:** Unlike opaque AI models, MediPrep utilizes an explicit, deterministic clinical rules engine that evaluates recorded vital signs against evidence-based perioperative safety thresholds:
  • Blood Sugar Rule: Blood sugar > 200 mg/dL triggers a High-Severity Alert: 'CLINICAL REVIEW REQUIRED: Severe Hyperglycemia detected.'
  • Hemoglobin Rule: Hemoglobin < 10 g/dL triggers a Warning: 'CLINICAL REVIEW REQUIRED: Low Hemoglobin (<10 g/dL) detected.'
  • Blood Pressure Rule: Systolic BP > 140 mmHg or Diastolic BP > 90 mmHg triggers: 'Elevated Blood Pressure detected.'
  • Body Temperature Rule: Temperature > 100.4°F triggers: 'Fever detected: Potential active infection contraindicating elective surgery.' Temperature < 95.0°F triggers: 'Hypothermia warning.'
  • Serum Creatinine Rule: Creatinine > 1.5 mg/dL triggers: 'Elevated Creatinine: Risk of renal impairment under procedural contrast/anesthesia.'

4.1.6 Procedure Readiness Evaluation & Real-Time Scoring

The Procedure Readiness Evaluation Engine (tbl_procedure_readiness) acts as the central intelligence of MediPrep, synthesizing multidisciplinary clinical data into an instantaneous, objective procedural clearance score.

- **Holistic Multi-Factor Evaluation:** The engine continuously monitors four foundational preparation pillars: (1) Checklist Item Completion: 100% completion of critical and routine

checklist tasks; (2) Report Verification: Every document upload task has an uploaded PDF verified by the attending doctor; (3) Assessment & Consent: Patient pre-op assessment is locked, and digital informed consent is confirmed by the patient and verified by the doctor; (4) Clinical Safety Rules: Zero unreviewed clinical vitals alerts or contraindications.

- **Color-Coded Readiness Classification:** The readiness status is dynamically classified and displayed across all stakeholder consoles:
  • GREEN (Ready for Procedure): All checklist tasks completed, reports verified, consent confirmed, and vitals normal. Patient is cleared for the OR.
  • YELLOW (Pending Doctor Review): Non-critical tasks pending, or vital signs flagged for doctor clinical review/override.
  • RED (Not Ready / Incomplete): One or more critical tasks incomplete, missing mandatory lab reports, unconfirmed consent, or severe contraindication.

4.1.7 Automated Notifications, Risk Alerts & Audit Logging

To ensure continuous situational awareness and legal accountability, MediPrep incorporates a dedicated communication and audit subsystem.

- **Real-Time System Notifications:** The notification engine (tbl_notifications) dispatches contextual alerts to users regarding procedure scheduling, upcoming checklist deadlines, new laboratory report uploads, and clarification responses.
- **Clinical Risk Alerts:** When the rules engine detects an abnormal metric, it generates a prioritized risk alert (tbl_risk_alerts) tagged with severity (Low, Medium, High). Alerts must be formally reviewed and acknowledged by the physician before procedure clearance can proceed.
- **Comprehensive Audit Logging:** To comply with healthcare regulatory standards (NABH/HIPAA), tbl_audit_logs captures an immutable record of every action: user logins, report uploads, doctor verifications, vitals changes, checklist completions, and readiness overrides, storing actor ID, role, action, details, and exact timestamps.

CHAPTER 5

FLOW DIAGRAMS

5.1 DATA FLOW DIAGRAMS (DFD)

A Data Flow Diagram (DFD) is a fundamental graphical modeling tool used in structured software engineering to visualize the flow of data through an information system. It depicts the transformation of inputs into outputs through sequential processes, highlighting data stores, external entities, and data pathways. The following sections outline the standard symbols, Context Level Diagram (Level 0), detailed Level 1 DFDs for each user role, and the comprehensive system architecture workflow.

5.1.1 DFD Symbols and Notations

MediPrep utilizes standard Gane and Sarson / Yourdon DFD notations as defined in Table 5.1:

Table 5.1: Standard Data Flow Diagram Symbols and Notations

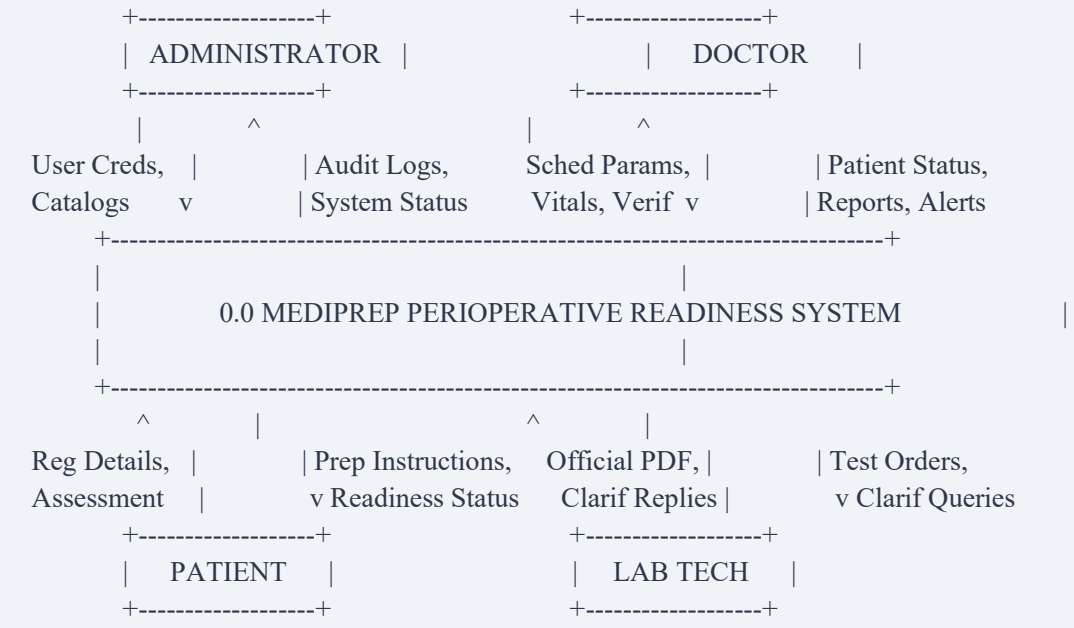
SYMBOL NAME	NOTATION / SHAPE	MEANING & FUNCTIONAL DESCRIPTION
External Entity	Rectangle / Box	Represents an external person, role, or system that injects data into or receives data from the system (e.g., Admin, Doctor, Patient, Lab Tech).
Process	Rounded Rectangle / Circle	Represents a business logic transformation or computation that manipulates data (e.g., 2.0 Procedure Scheduling, 5.0 Readiness Evaluation).
Data Store	Open-Ended Rectangle / Parallel Lines	Represents a persistent storage repository where data is retained for future retrieval (e.g., tbl_users, tbl_scheduled_procedures).
Data Flow	Directed Arrow (-->)	Represents the directional movement of packets of data between entities, processes, and data stores.

5.1.2 Level 0 DFD (Context Level Diagram)

The Level 0 Context Diagram establishes the boundary of the entire MediPrep platform. The central system (Process 0.0: MediPrep System) interfaces directly with four external entities:
• Administrator: Injects user registrations, role configurations, procedure catalog entries, and template definitions; receives audit logs and system activity metrics.
• Doctor: Injects procedure scheduling parameters, clinical vitals, report verifications, and clarification queries; receives patient checklists, lab reports, risk alerts, and readiness status.
• Patient: Injects registration details, pre-op health assessment responses, and consent confirmations; receives scheduled procedure details, prep instructions, and readiness progress.

• Hospital Laboratory Technician: Injects official PDF diagnostic reports and clarification replies; receives pending investigation test orders.

Figure 5.1: Context Level DFD (Level 0)



5.1.3 Level 1 DFDs

Level 1 Data Flow Diagrams decompose the central system into detailed operational sub-processes corresponding to the core user roles.

5.1.3.1 Admin Module DFD (Subsystem 1.0)

Process 1.1 manages user accounts and profiles, validating credentials and persisting records in tbl_users, tbl_doctor_profiles, and tbl_patient_profiles. Process 1.2 manages procedure definitions and dynamic checklist templates, persisting in tbl_procedures and tbl_dynamic_checklist_templates. Process 1.3 queries tbl_audit_logs to generate security reports for the Administrator.

5.1.3.2 Doctor Module DFD (Subsystem 2.0)

Process 2.1 receives patient selection and procedure parameters from the Doctor, writing records to tbl_scheduled_procedures and instantiating items in tbl_checklist_items. Process 2.2 reads uploaded reports from tbl_uploaded_reports, allows the Doctor to inspect PDFs, and records verification timestamps or initiates queries in tbl_lab_clarifications. Process 2.3 captures clinical vitals, writes to tbl_patient_vitals, and triggers the rules engine.

5.1.3.3 Patient Module DFD (Subsystem 3.0)

Process 3.1 reads scheduled procedures from tbl_scheduled_procedures and displays personalized preparation instructions and checklist tasks. Process 3.2 captures patient responses for the pre-procedure assessment, writes to tbl_patient_assessments, and locks the form. Process 3.3 presents the procedure-specific consent document, captures the patient's confirmation, and updates tbl_patient_consents.

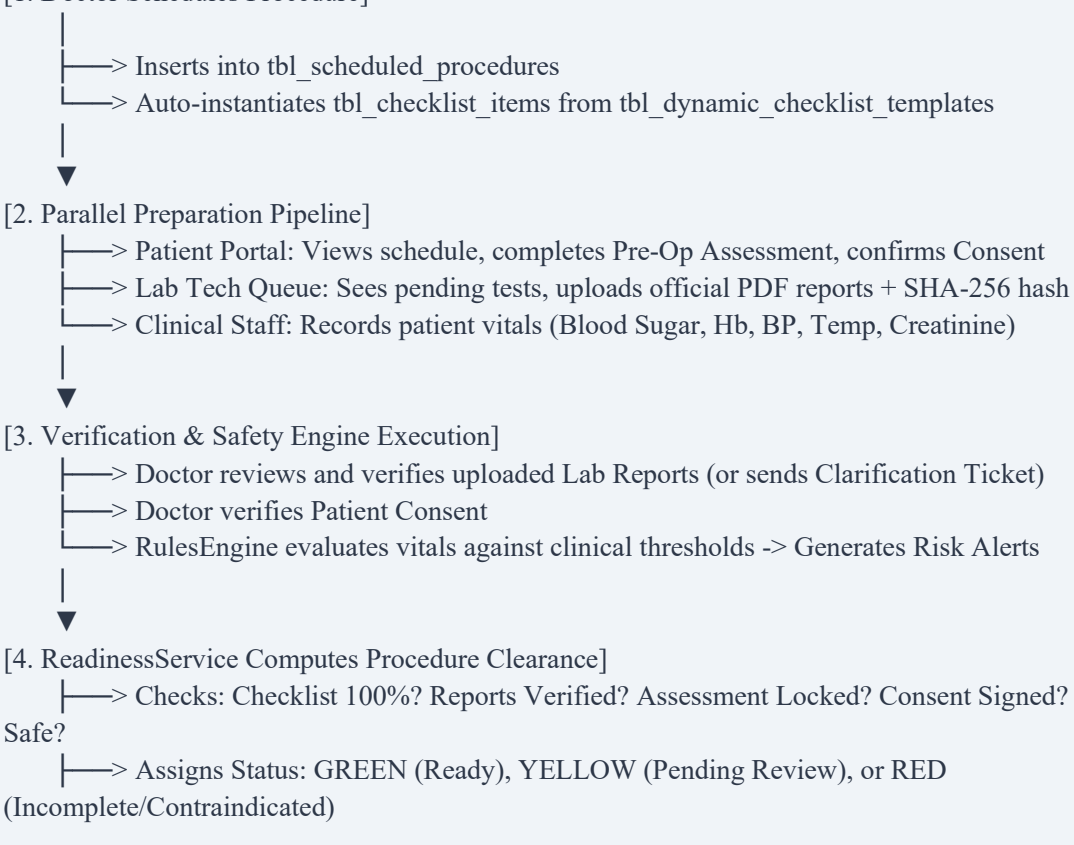
5.1.3.4 Hospital Laboratory Technician Module DFD (Subsystem 4.0)

Process 4.1 queries tbl_checklist_items for pending document upload tasks. Process 4.2 accepts official PDF reports, computes SHA-256 hashes, stores physical files in the uploads/ directory, and records metadata in tbl_uploaded_reports. Process 4.3 manages clarification tickets in tbl_lab_clarifications, allowing technicians to resolve inquiries submitted by doctors.

5.1.4 System Architecture & Data Flow Workflow

Figure 5.6 details the end-to-end data flow pipeline that unifies scheduling, task management, laboratory ingestion, rules evaluation, and readiness scoring into a synchronized operational lifecycle:

Figure 5.6: End-to-End Perioperative Clinical Workflow



> └──> Writes to tbl_procedure_readiness & records event in tbl_audit_logs

CHAPTER 6

SYSTEM DESIGN

6.1 SYSTEM DESIGN PRINCIPLES

System design translates the functional and non-functional requirements established in the SRS into a coherent, robust, and extensible software architecture. MediPrep adheres to core software engineering design principles: separation of concerns via the MVC paradigm, defense-in-depth security, strict data normalization (Third Normal Form - 3NF), and high user-experience usability. The following sections present the detailed Input Design, Output Design, and comprehensive Database Table Design.

6.2 INPUT DESIGN

Input design governs the specification of data entry forms, user controls, boundary validations, and error handling mechanisms to guarantee that only sanitized, accurate, and authorized data enters the system. The key input interfaces in MediPrep include:

6.2.1 Patient Registration Form

- **Username & Name Fields:** Captures alphanumeric username (unique), first name, and last name. Allows distinct users to share identical legal names while maintaining strict username uniqueness.
- **Date of Birth (DOB):** HTML5 date picker enforced with strict JavaScript and backend validation: DOB must be earlier than the current date. Future dates and today's date are rejected with clear error prompts.
- **Indian Mobile Number:** Text input validated via regex: must contain exactly 10 numeric digits, with the leading digit restricted to 6, 7, 8, or 9. Short codes and international formats without Indian prefixes are rejected.
- **Password & Confirmation:** Password fields enforced with minimum 8 characters rule; spaces are prohibited. Client-side and server-side matching validation.
- **Blood Group Dropdown:** Standardized select control containing: Select Blood Group, A+, A-, B+, B-, AB+, AB-, O+, O-, Not Known.

6.2.2 Doctor Procedure Scheduling Form

- **Patient Selection:** Dynamic dropdown populated exclusively with registered patient accounts.
- **Procedure Selection:** Dropdown linked to tbl_procedures master catalog.
- **Scheduled Date & Time:** Datetime-local control enforcing scheduled dates in the future.
- **Room Allocations:** Select inputs for Preparation Room and Operating Theater.

6.2.3 Hospital Lab Technician Report Upload Form

- **Investigation Target:** Auto-linked to the specific pending checklist item ID.
- **Laboratory Origin:** Dropdown selecting 'Central Hospital Clinical Laboratory' or accredited external partner 'ABC Diagnostics'.

- **External Report ID & Test Date:** Text and date controls capturing laboratory requisition numbers.
- **File Attachment:** File input strictly restricted to .pdf format; server computes SHA-256 hash upon upload.

6.2.4 Clinical Vitals Entry Form

- **Physiological Metrics:** Number fields with step increments capturing: Blood Sugar (mg/dL), Hemoglobin (g/dL), Systolic BP (mmHg), Diastolic BP (mmHg), Temperature (°F), and Serum Creatinine (mg/dL).

6.3 OUTPUT DESIGN

Output design ensures that processed data is communicated to healthcare stakeholders in an intelligible, actionable, and visually coherent format. MediPrep provides dedicated output consoles:

- **Patient Pre-Operative Preparation Portal:** Renders scheduled procedure details, countdown to procedure date/time, procedure room location, categorized preparation checklist with interactive checkboxes, assessment completion badge, and consent status.
- **Doctor Clinical & Verification Console:** Presents an active patient roster with real-time readiness status badges (Green/Yellow/Red), side-by-side PDF report viewer with 'Verify' and 'Request Clarification' action buttons, vitals review card, and one-click procedure clearance.
- **Laboratory Technician Investigation Queue:** Displays pending laboratory tests across all scheduled surgical patients, indicating patient name, test type, due date, and upload status.
- **Procedure Readiness Status Card:** A high-visibility card displaying overall procedure readiness (Green, Yellow, Red), compliance percentage progress bar, summary of completed vs pending tasks, and prioritized clinical risk alert banners.
- **System Audit Log Inspector:** An administrative tabular view displaying chronological system activity with actor ID, role, action, details, and timestamps.

6.4 DATABASE TABLE DESIGN (DATA DICTIONARY)

The MediPrep database (hospital_readiness_db) is normalized to Third Normal Form (3NF) to guarantee transactional integrity, eliminate data redundancy, and maintain strict referential integrity. The schema comprises exactly 20 tables managed by the MySQL InnoDB engine. Tables 6.1 through 6.20 present the complete data dictionary specifying column names, data types, lengths, nullability, key constraints, and functional descriptions.

Table 6.1: Database Table Design: tbl_roles (User Roles Master)

FIELD NAME	DATA TYPE & SIZE	NULL	KEY	DESCRIPTION & CONSTRAINTS

FIELD NAME	DATA TYPE & SIZE	NULL	KEY	DESCRIPTION & CONSTRAINTS
id	INT(11)	NO	PRI	Primary Key, Auto Increment
name	VARCHAR(50)	NO	UNI	Unique role name (admin, doctor, patient, lab_tech)
description	VARCHAR(255)	YES		Descriptive text of role privileges

Table 6.2: Database Table Design: tbl_users (System Users Account Store)

FIELD NAME	DATA TYPE & SIZE	NULL	KEY	DESCRIPTION & CONSTRAINTS
id	INT(11)	NO	PRI	Primary Key, Auto Increment
role_id	INT(11)	NO	MUL	Foreign Key referencing tbl_roles(id)
username	VARCHAR(50)	NO	UNI	Unique login identifier
email	VARCHAR(100)	NO	UNI	Unique email address
password_hash	VARCHAR(255)	NO		PBKDF2/SHA-256 encrypted password hash
first_name	VARCHAR(50)	NO		Legal first name of user
last_name	VARCHAR(50)	NO		Legal last name of user
phone	VARCHAR(20)	YES		10-digit Indian mobile contact number
is_active	TINYINT(1)	NO		Account status flag (1=active, 0=disabled)
created_at	TIMESTAMP	NO		Record creation timestamp
updated_at	TIMESTAMP	NO		Record modification timestamp

Table 6.3: Database Table Design: tbl_doctor_profiles (Doctor Profile Details)

FIELD NAME	DATA TYPE & SIZE	NULL	KEY	DESCRIPTION & CONSTRAINTS
id	INT(11)	NO	PRI	Primary Key, Auto Increment
user_id	INT(11)	NO	UNI	Foreign Key referencing tbl_users(id)
license_number	VARCHAR(100)	NO	UNI	Unique medical council registration number
specialty	VARCHAR(100)	NO		Clinical medical specialty (Cardiology, Surgery)
department	VARCHAR(100)	NO		Hospital department designation

Table 6.4: Database Table Design: tbl_patient_profiles (Patient Demographic Details)

FIELD NAME	DATA TYPE & SIZE	NULL	KEY	DESCRIPTION & CONSTRAINTS
id	INT(11)	NO	PRI	Primary Key, Auto Increment
user_id	INT(11)	NO	UNI	Foreign Key referencing tbl_users(id)
date_of_birth	DATE	NO		Date of birth (validated strictly in past)
gender	ENUM	NO		Gender: 'male', 'female', 'other'
blood_type	VARCHAR(20)	YES		Blood group (A+, B+, O+, AB+, etc.)
medical_history	TEXT	YES		Baseline chronic conditions and notes
emergency_contact_name	VARCHAR(100)	NO		Name of emergency contact person
emergency_contact_phone	VARCHAR(20)	NO		Emergency contact telephone number

Table 6.5: Database Table Design: tbl_lab_profiles (Laboratory Profiles)

FIELD NAME	DATA TYPE & SIZE	NULL	KEY	DESCRIPTION & CONSTRAINTS
id	INT(11)	NO	PRI	Primary Key, Auto Increment
user_id	INT(11)	NO	UNI	Foreign Key referencing tbl_users(id)
lab_name	VARCHAR(150)	NO		Accredited name of the clinical laboratory
lab_type	ENUM	NO		Type: 'hospital' or 'external'
license_number	VARCHAR(100)	YES		Clinical laboratory license identifier
accreditation	VARCHAR(100)	YES		Accreditation body (NABL, CAP, ISO)
created_at	TIMESTAMP	NO		Profile registration timestamp

Table 6.6: Database Table Design: tbl_procedures (Procedure Master Catalog)

FIELD NAME	DATA TYPE & SIZE	NULL	KEY	DESCRIPTION & CONSTRAINTS
id	INT(11)	NO	PRI	Primary Key, Auto Increment
code	VARCHAR(20)	NO	UNI	Unique procedure code (e.g., PROC-LHC-01)
name	VARCHAR(100)	NO	MUL	Procedure name (e.g., Left Heart Catheterization)
description	TEXT	YES		Detailed clinical explanation of intervention
category	VARCHAR(50)	NO		Specialty category (Cardiology, Gastroenterology)
est_duration_mins	INT(11)	NO		Estimated operative duration in minutes

Table 6.7: Database Table Design: tbl_procedure_rooms (Procedure Room Master)

FIELD NAME	DATA TYPE & SIZE	NULL	KEY	DESCRIPTION & CONSTRAINTS
id	INT(11)	NO	PRI	Primary Key, Auto Increment
procedure_id	INT(11)	NO	MUL	Foreign Key referencing tbl_procedures(id)
room_type	ENUM	NO		Room classification: 'preparation' or 'procedure'
room_name	VARCHAR(100)	NO		Room designation (e.g., Cath Lab 2, Prep Room A)

Table 6.8: Database Table Design: tbl_dynamic_checklist_templates (Checklist Master Templates)

FIELD NAME	DATA TYPE & SIZE	NULL	KEY	DESCRIPTION & CONSTRAINTS
id	INT(11)	NO	PRI	Primary Key, Auto Increment
procedure_id	INT(11)	NO	MUL	Foreign Key referencing tbl_procedures(id)
task_name	VARCHAR(100)	NO		Task designation (e.g., Fasting NPO 8 Hours)
task_type	ENUM	NO		Type: 'patient_action', 'staff_verification', 'document_upload'
description	TEXT	YES		Detailed task execution guidelines
due_days_before	INT(11)	NO		Target due deadline in days prior to procedure
is_critical	TINYINT(1)	NO		Criticality flag (1=blocking, 0=routine)

Table 6.9: Database Table Design: tbl_scheduled_procedures (Scheduled Procedures Registry)

FIELD NAME	DATA TYPE & SIZE	NULL	KEY	DESCRIPTION & CONSTRAINTS
id	INT(11)	NO	PRI	Primary Key, Auto Increment
patient_id	INT(11)	NO	MUL	Foreign Key referencing tbl_users(id)
doctor_id	INT(11)	NO	MUL	Foreign Key referencing tbl_users(id)
procedure_id	INT(11)	NO	MUL	Foreign Key referencing tbl_procedures(id)
scheduled_at	DATETIME	NO	MUL	Target date and time of procedure execution
preparation_room	VARCHAR(100)	YES		Assigned preparation holding bay
procedure_room	VARCHAR(100)	YES		Assigned operative theater / cath lab
location	VARCHAR(100)	NO		Hospital facility location/wing
status	ENUM	NO		Status: 'scheduled', 'completed', 'cancelled', 'delayed'
check_in_status	ENUM	NO		Check-in: 'scheduled', 'arrived', 'checked_in', 'ready_for_procedure'
created_at	TIMESTAMP	NO		Booking creation timestamp

Table 6.10: Database Table Design: tbl_checklist_items (Patient Specific Checklist Items)

FIELD NAME	DATA TYPE & SIZE	NULL	KEY	DESCRIPTION & CONSTRAINTS
id	INT(11)	NO	PRI	Primary Key, Auto Increment
scheduled_procedure_id	INT(11)	NO	MUL	FK referencing tbl_scheduled_procedures(id)

FIELD NAME	DATA TYPE & SIZE	NULL	KEY	DESCRIPTION & CONSTRAINTS
task_name	VARCHAR(100)	NO		Task designation
task_type	ENUM	NO		Type: 'patient_action', 'staff_verification', 'document_upload'
description	TEXT	YES		Detailed instructions for patient or staff
due_by_datetime	DATETIME	NO	MUL	Calculated absolute deadline timestamp
is_critical	TINYINT(1)	NO		Critical flag (1=blocks readiness, 0=non-blocking)
is_external	TINYINT(1)	NO		Flag indicating test conducted at external lab
lab_name	VARCHAR(150)	YES		Designated laboratory facility name
investigation_type	VARCHAR(100)	YES		Test type (CBC, Blood Sugar, ECG, Renal)
status	ENUM	NO		Status: 'pending', 'completed', 'verified', 'failed'
completed_by_user_id	INT(11)	YES	MUL	User ID of person completing task
completed_at	DATETIME	YES		Completion timestamp
verified_by_user_id	INT(11)	YES	MUL	Doctor user ID verifying task
verified_at	DATETIME	YES		Doctor verification timestamp
notes	TEXT	YES		Clinical observation and execution notes

Table 6.11: Database Table Design: tbl_uploaded_reports (Official Laboratory Reports Store)

FIELD NAME	DATA TYPE & SIZE	NULL	KEY	DESCRIPTION & CONSTRAINTS
id	INT(11)	NO	PRI	Primary Key, Auto Increment
checklist_item_id	INT(11)	NO	MUL	FK referencing tbl_checklist_items(id)
uploaded_by_user_id	INT(11)	YES		FK referencing tbl_users(id) (Lab Tech)
lab_name	VARCHAR(150)	YES		Name of performing laboratory
lab_type	ENUM	NO		Origin: 'hospital' or 'external'
external_report_id	VARCHAR(100)	YES		External laboratory requisition identifier
report_date	DATE	YES		Date specimen collected/analyzed
verification_status	ENUM	NO		'pending_verification', 'verified', 'needs_review', 'needs_clarification', 'external_received'
verification_notes	TEXT	YES		Doctor's clinical verification remarks
verified_by_doctor_id	INT(11)	YES		FK referencing tbl_users(id) (Doctor)
verified_at	DATETIME	YES		Timestamp of formal doctor verification
is_locked	TINYINT(1)	NO		Immutability lock (1=locked, 0=editable)
version	INT(11)	NO		Report version counter (defaults to 1)
is_current	TINYINT(1)	NO		Active version flag (1=current active version)

FIELD NAME	DATA TYPE & SIZE	NULL	KEY	DESCRIPTION & CONSTRAINTS
file_hash	VARCHAR(64)	YES		SHA-256 cryptographic file checksum
file_name	VARCHAR(255)	NO		Original uploaded filename
file_path	VARCHAR(512)	NO		Physical filesystem storage path
file_size_bytes	INT(11)	NO		Document file size in bytes
uploaded_at	TIMESTAMP	NO		Upload creation timestamp

Table 6.12: Database Table Design: tbl_lab_clarifications (Doctor-Lab Clarification Tickets)

FIELD NAME	DATA TYPE & SIZE	NULL	KEY	DESCRIPTION & CONSTRAINTS
id	INT(11)	NO	PRI	Primary Key, Auto Increment
scheduled_procedure_id	INT(11)	NO	MUL	FK referencing tbl_scheduled_procedures(id)
checklist_item_id	INT(11)	YES		FK referencing tbl_checklist_items(id)
doctor_id	INT(11)	NO		FK referencing tbl_users(id) (Doctor)
lab_user_id	INT(11)	YES		FK referencing tbl_users(id) (Lab Tech)
request_message	TEXT	NO		Doctor's clinical inquiry regarding report
response_message	TEXT	YES		Technician's clarification response
status	ENUM	NO		Ticket state: 'pending', 'responded', 'resolved'
created_at	TIMESTAMP	NO		Inquiry creation timestamp

FIELD NAME	DATA TYPE & SIZE	NULL	KEY	DESCRIPTION & CONSTRAINTS
updated_at	TIMESTAMP	NO		Last modification timestamp

Table 6.13: Database Table Design: tbl_patient_vitals (Pre-Op Physiological Vitals)

FIELD NAME	DATA TYPE & SIZE	NULL	KEY	DESCRIPTION & CONSTRAINTS
id	INT(11)	NO	PRI	Primary Key, Auto Increment
scheduled_procedure_id	INT(11)	NO	UNI	FK referencing tbl_scheduled_procedures(id)
blood_sugar	DECIMAL(5,2)	YES		Fasting/random glucose in mg/dL (>200 triggers alert)
hemoglobin	DECIMAL(4,2)	YES		Hemoglobin concentration in g/dL (<10 triggers alert)
systolic_bp	INT(11)	YES		Systolic blood pressure in mmHg (>140 triggers alert)
diastolic_bp	INT(11)	YES		Diastolic blood pressure in mmHg (>90 triggers alert)
temperature	DECIMAL(4,1)	YES		Body temperature in °F (>100.4 or <95 triggers alert)
creatinine	DECIMAL(4,2)	YES		Serum creatinine in mg/dL (>1.5 triggers alert)
recorded_at	TIMESTAMP	NO		Vitals recording timestamp

Table 6.14: Database Table Design: tbl_patient_assessments (Pre-Op Health Questionnaire)

FIELD NAME	DATA TYPE & SIZE	NULL	KEY	DESCRIPTION & CONSTRAINTS
id	INT(11)	NO	PRI	Primary Key, Auto Increment

FIELD NAME	DATA TYPE & SIZE	NULL	KEY	DESCRIPTION & CONSTRAINTS
scheduled_procedure_id	INT(11)	NO	UNI	FK referencing tbl_scheduled_procedures(id)
allergies	TEXT	YES		Known drug, latex, or food allergies
current_medications	TEXT	YES		Active prescription and OTC medications
previous_surgeries	TEXT	YES		Historical surgical procedures
anesthesia_problems	TEXT	YES		Past adverse anesthesia reactions
medical_history_notes	TEXT	YES		General systemic medical conditions
emergency_contact_name	VARCHAR(100)	YES		Emergency contact person name
emergency_contact_phone	VARCHAR(20)	YES		Emergency contact phone number
status	ENUM	NO		Status: 'draft' or 'submitted_locked'
submitted_at	DATETIME	YES		Locking submission timestamp
created_at	TIMESTAMP	NO		Record creation timestamp

Table 6.15: Database Table Design: tbl_patient_consents (Informed Procedural Consents)

FIELD NAME	DATA TYPE & SIZE	NULL	KEY	DESCRIPTION & CONSTRAINTS
id	INT(11)	NO	PRI	Primary Key, Auto Increment
scheduled_procedure_id	INT(11)	NO	MUL	FK referencing tbl_scheduled_procedures(id)
consent_text	TEXT	NO		Legal text of procedural risks and authorization

FIELD NAME	DATA TYPE & SIZE	NULL	KEY	DESCRIPTION & CONSTRAINTS
version	INT(11)	NO		Consent document version number
is_current	TINYINT(1)	NO		Active version flag (1=active)
status	ENUM	NO		State: 'generated', 'confirmed_locked', 'verified'
confirmed_by_patient_id	INT(11)	YES		FK referencing tbl_users(id) (Patient)
confirmed_at	DATETIME	YES		Patient digital confirmation timestamp
verified_by_doctor_id	INT(11)	YES		FK referencing tbl_users(id) (Doctor)
verified_at	DATETIME	YES		Doctor legal verification timestamp
created_at	TIMESTAMP	NO		Consent generation timestamp

Table 6.16: Database Table Design: tbl_procedure_readiness (Procedure Clearance Status)

FIELD NAME	DATA TYPE & SIZE	NULL	KEY	DESCRIPTION & CONSTRAINTS
id	INT(11)	NO	PRI	Primary Key, Auto Increment
scheduled_procedure_id	INT(11)	NO	UNI	FK referencing tbl_scheduled_procedures(id)
readiness_status	ENUM	NO		Score: 'red', 'orange', 'yellow', 'green'
evaluated_by_user_id	INT(11)	YES	MUL	FK referencing tbl_users(id) (evaluating user)
last_evaluated_at	TIMESTAMP	NO		Evaluation execution timestamp
notes	TEXT	YES		Readiness breakdown and compliance remarks

FIELD NAME	DATA TYPE & SIZE	NULL	KEY	DESCRIPTION & CONSTRAINTS
override_reason	TEXT	YES		Doctor justification for manual clinical override

Table 6.17: Database Table Design: tbl_risk_alerts (Clinical Contraindication Alerts)

FIELD NAME	DATA TYPE & SIZE	NULL	KEY	DESCRIPTION & CONSTRAINTS
id	INT(11)	NO	PRI	Primary Key, Auto Increment
scheduled_procedure_id	INT(11)	NO	MUL	FK referencing tbl_scheduled_procedures(id)
alert_type	VARCHAR(100)	NO		Classification (e.g., Blood Sugar, BP, Fever)
alert_message	TEXT	NO		Clinical warning message and threshold breach
severity	ENUM	NO		Risk level: 'low', 'medium', 'high'
is_resolved	TINYINT(1)	NO	MUL	Resolution flag (1=resolved, 0=active)
resolved_by_user_id	INT(11)	YES	MUL	FK referencing tbl_users(id) (Doctor)
resolved_at	DATETIME	YES		Doctor resolution timestamp
created_at	TIMESTAMP	NO		Alert trigger timestamp

Table 6.18: Database Table Design: tbl_health_suggestions (Pre-Op Health Advisory Rules)

FIELD NAME	DATA TYPE & SIZE	NULL	KEY	DESCRIPTION & CONSTRAINTS
id	INT(11)	NO	PRI	Primary Key, Auto Increment
procedure_id	INT(11)	NO	MUL	FK referencing tbl_procedures(id)

FIELD NAME	DATA TYPE & SIZE	NULL	KEY	DESCRIPTION & CONSTRAINTS
suggestion_text	TEXT	NO		Health advice text displayed to patient
trigger_condition	VARCHAR(255)	NO		Condition triggering advisory display
created_at	TIMESTAMP	NO		Creation timestamp

Table 6.19: Database Table Design: tbl_notifications (User Alert Notifications)

FIELD NAME	DATA TYPE & SIZE	NULL	KEY	DESCRIPTION & CONSTRAINTS
id	INT(11)	NO	PRI	Primary Key, Auto Increment
user_id	INT(11)	NO	MUL	Recipient user FK referencing tbl_users(id)
title	VARCHAR(150)	NO		Notification header title
message	TEXT	NO		Notification message body
type	ENUM	NO		Channel: 'sms', 'email', 'in_app'
status	ENUM	NO	MUL	Delivery status: 'unread', 'read', 'failed'
sent_at	TIMESTAMP	NO		Dispatch timestamp
sender_id	INT(11)	YES	MUL	Sender FK referencing tbl_users(id)
scheduled_procedure_id	INT(11)	YES	MUL	FK referencing tbl_scheduled_procedures(id)

Table 6.20: Database Table Design: tbl_audit_logs (Immutable Security Audit Trail)

FIELD NAME	DATA TYPE & SIZE	NULL	KEY	DESCRIPTION & CONSTRAINTS

FIELD NAME	DATA TYPE & SIZE	NULL	KEY	DESCRIPTION & CONSTRAINTS
id	INT(11)	NO	PRI	Primary Key, Auto Increment
user_id	INT(11)	YES		Actor FK referencing tbl_users(id)
role_name	VARCHAR(50)	YES		Role name of actor at event execution
action	VARCHAR(100)	NO		Standardized action verb (e.g., PROCEDURE_SCHEDULED)
details	TEXT	YES		Full JSON or textual description of transaction parameters
patient_id	INT(11)	YES		FK referencing affected patient user ID
scheduled_procedure_id	INT(11)	YES		FK referencing affected scheduled procedure
created_at	TIMESTAMP	NO		Immutable event timestamp

CHAPTER 7

TOOLS AND PLATFORM

7.1 FRONT END TECHNOLOGIES

The front-end tier of MediPrep is engineered to deliver a responsive, accessible, and intuitive user experience across desktop workstations, clinical tablet devices, and patient mobile smartphones without requiring proprietary client-side installations. The core client technologies include:

- **HTML5 (HyperText Markup Language 5):** Provides semantic structural architecture across all application views. Form inputs leverage specialized HTML5 input types (e.g., date, time, number, email, and file upload) with native pattern constraints and accessibility attributes.
- **CSS3 (Cascading Style Sheets 3):** Delivers modern, responsive visual styling utilizing CSS Flexbox and Grid layouts, CSS custom properties (variables) for consistent theming, smooth transitions, and high-contrast color palettes tailored for clinical readability.
- **JavaScript (ECMAScript 6+):** Executes client-side interactivity, asynchronous DOM updates, dynamic form validation, and real-time input sanitization. It prevents invalid form submissions—such as future dates of birth or improperly formatted mobile numbers—prior to network dispatch.
- **Bootstrap 5 Framework:** Implements a modern 12-column responsive grid system, comprehensive UI components (navbars, modals, progress bars, accordion checklist containers, and alert banners), and utilities that ensure cross-browser compatibility across Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari.

7.2 BACK END TECHNOLOGIES

The back-end server architecture is constructed using modern, high-performance, open-source technologies designed for rapid execution and rock-solid security:

- **Python 3.10+:** Serves as the primary server-side programming language. Python offers exceptional readability, strict dynamic typing, robust exception handling, and a comprehensive standard library including hashlib (for SHA-256 report verification), datetime (for strict scheduling arithmetic), and logging (for security audit trails).
- **Flask Micro-Framework (Version 3.x):** A lightweight, modular WSGI web application framework. Flask provides high routing flexibility through Blueprint modularization, built-in request/response abstraction, session cookie management, and minimal runtime overhead, making it ideal for healthcare microservices.
- **Jinja2 Templating Engine:** Enables powerful server-side template rendering, template inheritance (base layout extensions), context processors for injecting active user session details, and automated HTML escaping to neutralize Cross-Site Scripting (XSS) attack vectors.
- **Werkzeug Security Primitives:** Provides industrial-strength cryptographic password hashing using the PBKDF2 algorithm with HMAC-SHA256, secure token generation, and

secure filename sanitization (secure_filename) to thwart directory traversal attacks during report uploads.

7.3 FRAMEWORK & ARCHITECTURAL PATTERN

MediPrep strictly implements the 3-Tier Model-View-Controller (MVC) architectural pattern augmented with the Service-Repository design pattern. This architecture ensures complete separation of concerns:
1. Models & Repositories (Data Access Layer): Abstract raw SQL operations. Repositories (e.g., UserRepository, DoctorRepository, PatientRepository, LabRepository, AuditRepository) execute parameterized queries using connection context managers, ensuring total protection against SQL injection.
2. Services Layer (Business Logic): Encapsulates domain logic independently of HTTP protocols. Services (e.g., ReadinessService, RulesEngine, DoctorService, LabService) execute complex computations such as clinical vital evaluations and readiness scoring.
3. Controllers / Routes (Presentation Layer): Flask Blueprints (auth, admin, doctor, patient, lab) handle HTTP requests, validate input, invoke domain services, and return rendered Jinja2 HTML views or JSON payloads.

7.4 DEVELOPMENT PLATFORM & OPERATING ENVIRONMENT

- **Operating System:** Microsoft Windows 10/11 Professional (64-bit) development environment; fully compatible with enterprise Linux servers (Ubuntu 22.04 LTS / Rocky Linux) for production deployment.
- **Development Environment:** Visual Studio Code (VS Code) equipped with Python IntelliSense, Pylance, Jinja syntax highlighting, and integrated terminal.
- **Version Control System:** Git 2.4x for distributed version control, feature branching, and atomic commit history tracking.
- **Local Server Stack:** XAMPP 8.2 providing the Apache HTTP Server and local MySQL / MariaDB database server.

7.5 RELATIONAL DATABASE MANAGEMENT SYSTEM

The persistence tier is powered by MySQL 8.0 / MariaDB. The InnoDB storage engine is enforced across all 20 tables to guarantee strict ACID (Atomicity, Consistency, Isolation, Durability) properties, foreign key referential integrity, row-level locking, and automated transaction rollbacks upon unexpected execution exceptions.

CHAPTER 8

TESTING

8.1 UNIT TESTING

Unit testing involves the verification of individual software components and functions in isolation to ensure they produce expected outputs for given inputs. In MediPrep, unit tests were systematically executed on:

• Cryptographic Password Hashing: Testing generate_password_hash and check_password_hash to verify that plain-text passwords are never persisted and hashes match successfully.

• Indian Mobile Regex Validation: Testing regex pattern ^[6-9]\d{9}$ against valid numbers (e.g., 9876543210, 8123456789) and invalid cases (e.g., 5551234, 1555, characters, short codes).

• Date of Birth Validation: Testing that dates in the past (e.g., 1990-05-15, 2026-10-08) return valid, while today's date and future dates return validation errors.

• RulesEngine Threshold Functions: Testing vitals evaluation against boundary conditions (Blood Sugar = 199 vs 201 mg/dL; Hemoglobin = 9.9 vs 10.1 g/dL; Systolic BP = 139 vs 141 mmHg; Temperature = 99.0°F vs 100.5°F; Creatinine = 1.4 vs 1.6 mg/dL).

8.2 INTEGRATION TESTING

Integration testing verifies the functional interoperability between interconnected modules and database repositories:

• Procedure Scheduling & Dynamic Checklist Generation: Verified that when a Doctor schedules a procedure, a record is created in tbl_scheduled_procedures and corresponding checklist items are automatically populated in tbl_checklist_items with correct due dates.

• Hospital Lab Technician Upload & File Hashing: Verified that uploading a valid PDF computes a matching SHA-256 hash, stores the file in uploads/, records metadata in tbl_uploaded_reports, and changes the checklist item status to pending verification.

• Two-Tier Verification Loop: Verified that Doctor report verification updates verification_status to 'verified', updates verified_at, and triggers ReadinessService recalculation.

• Session & Role Decorator Enforcement: Verified that unauthenticated users are redirected to login and authenticated users with the 'patient' role cannot access '/doctor' or '/admin' endpoints.

8.3 SYSTEM TESTING

System testing evaluates the end-to-end functionality of the completely integrated MediPrep platform against specified requirements.

8.3.1 Acceptance Testing

Acceptance testing validated complete user operational journeys: (1) Admin seeds procedures and templates; (2) Patient registers with valid DOB and mobile number; (3)

Doctor schedules 'Left Heart Catheterization'; (4) Patient logs in, sees scheduled procedure, completes Pre-Op Assessment, and signs Consent; (5) Lab Tech uploads CBC and ECG reports; (6) Doctor verifies reports and records normal vitals; (7) System clears patient to GREEN status.

8.3.2 Validation Testing

Validation testing tested input boundaries: rejecting duplicate usernames and emails without wiping out valid form entries; enforcing minimum 8-character passwords; ensuring mandatory clinical fields cannot be bypassed via direct HTTP POST requests.

8.3.3 Black Box Testing

Black box testing focused on Equivalence Partitioning and Boundary Value Analysis of functional requirements without inspecting underlying source code. Inputs were provided via browser UI forms, and resulting screens and alerts were compared against expected outputs.

8.3.4 White Box Testing

White box testing analyzed code paths, branch coverage, and cyclomatic complexity within RulesEngine.evaluate_vitals and ReadinessService.evaluate_and_update_readiness, ensuring every conditional branch (all normal, single breach, multiple breaches, missing vitals) was exercised and covered.

8.4 SAMPLE TEST CASES

Table 8.1 documents 16 formal test cases executed across the MediPrep platform, detailing test case identifiers, modules, test steps, expected results, actual outcomes, and verification status.

Table 8.1: Formal Test Cases and Execution Results (TC01 to TC16)

TC ID	MODULE	TEST DESCRIPTION & INPUT	EXPECTED RESULT	ACTUAL RESULT	STATUS
TC01	Auth	User registration with DOB = today / future date	Form rejected with 'DOB must be strictly in the past' error	Rejected with clear error message beside DOB field	PASS
TC02	Auth	Registration with past DOB (e.g., 08-10-2026 when today is 20-10-2026)	Form accepts date without age error	Accepted successfully without rejection	PASS

TC ID	MODULE	TEST DESCRIPTION & INPUT	EXPECTED RESULT	ACTUAL RESULT	STATUS
TC03	Auth	Registration with invalid mobile number '5551234' or '1555'	Form rejected: 'Enter a valid 10-digit Indian mobile number'	Rejected; error shown; other fields preserved	PASS
TC04	Auth	Registration with valid 10-digit mobile '9876543210'	Accepted as valid Indian mobile number	Accepted successfully	PASS
TC05	Auth	Registration with password < 8 characters (e.g., 'Pass1')	Form rejected: 'Password must contain at least 8 characters'	Rejected with prompt; submission blocked	PASS
TC06	Auth	Registration with duplicate username or email	Form rejected: 'Username/Email already exists'; preserves other fields	Rejected with error; form details preserved	PASS
TC07	Auth	RBAC Group: Dropdown contains Patient, Doctor, Lab Tech	Displays exactly the three allowed roles (no Admin or External Lab)	Dropdown contains exactly the 3 allowed roles	PASS
TC08	Doctor	Doctor schedules procedure for registered patient 'John Doe'	Schedule saved; linked to patient; checklist auto-generated	Procedure appears on Patient Dashboard with prep tasks	PASS
TC09	Patient	Patient completes pre-op assessment and submits	Assessment status set to 'submitted_locked'; cannot be edited	Assessment locked successfully; draft status closed	PASS
TC10	Lab Tech	Hospital Lab Tech uploads official PDF for ECG report	File saved in uploads/; SHA-256 hash computed; status = pending	Report recorded with hash; awaiting doctor review	PASS
TC11	Doctor	Doctor verifies uploaded ECG report in clinical console	Report verification_status updated to 'verified'; timestamp set	Status set to verified; verified_at recorded	PASS

TC ID	MODULE	TEST DESCRIPTION & INPUT	EXPECTED RESULT	ACTUAL RESULT	STATUS
TC12	Doctor	Doctor sends clarification ticket to Lab Tech	Clarification record created in tbl_lab_clarifications; status pending	Ticket created; visible on Lab Tech console	PASS
TC13	Rules	Vitals entered: Blood Sugar = 240 mg/dL (> 200 threshold)	High severity alert triggered: 'Severe Hyperglycemia detected'	Risk alert generated; readiness marked YELLOW/RED	PASS
TC14	Rules	Vitals entered: Systolic BP = 150 mmHg (> 140 threshold)	Clinical alert triggered: 'Elevated Blood Pressure detected'	Alert generated and displayed on doctor console	PASS
TC15	Readiness	All checklist tasks complete, reports verified, vitals normal	Readiness engine calculates overall status = GREEN (Ready)	Overall status displayed as GREEN on all dashboards	PASS
TC16	Audit	User performs login, scheduling, upload, and clearance	All actions recorded in tbl_audit_logs with timestamps and IDs	Audit records verified in database with exact details	PASS

CHAPTER 9

SYSTEM IMPLEMENTATION

9.1 IMPLEMENTATION METHODOLOGY & STRATEGY

System implementation encompasses the practical translation of the architectural designs and software components into a fully operational production environment. The implementation of MediPrep followed a phased deployment strategy:

- **Phase 1: Environment Provisioning & Database Seeding:** Configuring the Python 3.10 virtual environment, installing dependencies (Flask, PyMySQL, mysql-connector-python, python-docx), executing database schema definitions (schema.sql), and seeding initial master catalogs for roles, procedures, and checklist templates (seed_data.sql).
- **Phase 2: Core Domain Logic & Security Engine Implementation:** Developing repositories and service layers, embedding PBKDF2/SHA-256 password hashing, writing custom route decorators for RBAC enforcement, and coding the deterministic RulesEngine and ReadinessService algorithms.
- **Phase 3: Controller Blueprints & View Template Integration:** Constructing modular Flask Blueprints (auth, admin, doctor, patient, lab) and integrating responsive Bootstrap 5 Jinja2 templates with client-side JavaScript input boundary validation.
- **Phase 4: Laboratory Ingestion Pipeline & File Security:** Configuring the uploads/ directory with strict file permissions, implementing SHA-256 file hashing, and establishing the doctor verification loop.
- **Phase 5: Comprehensive Testing & System Hardening:** Executing all 16 formal test cases, verifying audit log capture, sanitizing error outputs, and verifying cross-browser responsiveness.

9.2 PROBLEM STATEMENT

In modern healthcare administration, hospital surgical suites and interventional procedure suites represent the single most resource-intensive and revenue-critical assets. However, operating room efficiency is severely impaired by preventable, same-day procedural cancellations, which affect between 15% and 22% of elective surgical schedules globally. The primary root cause is information fragmentation across the perioperative journey: pre-procedure fasting guidelines are unmonitored; required diagnostic laboratory tests are missing, delayed, or unverified; patient informed consents remain unsigned; and acute physiological contraindications (such as severe hyperglycemia, severe anemia, or hypertensive crises) are discovered only after the patient is transferred into the operative theater. This breakdown causes massive hospital financial losses, squanders surgical team time, increases patient morbidity, and imposes severe emotional distress.

9.3 PROBLEM DEFINITION

Formally, the pre-procedure readiness problem can be modeled as a multi-constraint satisfaction optimization problem. Let P represent a scheduled procedure for patient u scheduled at time T. Let C = {c1, c2, ..., cn} represent the set of mandatory preparation

checklist items, L = {l1, l2, ..., lm} represent required diagnostic laboratory reports, A represent the pre-operative health assessment, S represent signed informed consent, and V = {v_sugar, v_hb, v_bp_sys, v_bp_dia, v_temp, v_creat} represent recorded physiological vitals.

The overall procedural readiness function R(P) evaluates to GREEN (Cleared) if and only if:
1. ∀ c_i ∈ C: status(c_i) = 'completed'
2. ∀ l_j ∈ L: verification_status(l_j) = 'verified' ∧ verified_by_doctor_id is not null
3. status(A) = 'submitted_locked'
4. status(S) = 'confirmed_locked' ∧ verified_by_doctor_id is not null
5. RulesEngine(V) = ∅ (No active clinical contraindications)

If any mandatory condition in (1)-(4) is unfulfilled, R(P) evaluates to RED (Incomplete). If vitals indicate non-critical anomalies requiring clinical review, R(P) evaluates to YELLOW (Pending Review). MediPrep solves this multi-constraint satisfaction problem through deterministic software automation, eliminating administrative failure points.

CHAPTER 10

CONCLUSION AND FUTURE ENHANCEMENTS

10.1 CONCLUSION

The successful development and deployment of "MediPrep – Hospital Procedure Preparation and Readiness System" provides a transformative digital solution to the longstanding operational and clinical challenges of hospital perioperative management. By systematically replacing fragmented paper binders, untracked telephone calls, and unverified patient assertions with a unified, role-governed web platform, MediPrep delivers significant clinical and administrative benefits:

• Elimination of Preventable Cancellations: By continuously tracking checklist items and computing real-time readiness status (Green/Yellow/Red), surgical teams can identify and resolve preparation deficiencies days before the procedure, eliminating morning-of-surgery cancellations.
• Guaranteed Report Provenance & Authenticity: Prohibiting direct patient uploads and enforcing authorized Hospital Laboratory Technician uploads with SHA-256 file hashing guarantees that only authentic diagnostic reports reach the attending physician.
• Deterministic Patient Safety: The automated Clinical Safety Rules Engine acts as an unwavering clinical safety net, flagging hyperglycemia, severe anemia, hypertension, fever, and renal impairment before anesthesia administration.
• Legal and Clinical Accountability: The immutable audit logging subsystem records every user action, ensuring full compliance with healthcare accreditation and regulatory standards.

In conclusion, MediPrep demonstrates that modern open-source web technologies—Python, Flask, and MySQL—when coupled with disciplined MVC architecture and deterministic clinical logic, can dramatically enhance hospital operating room efficiency, optimize healthcare economics, and safeguard patient clinical outcomes.

10.2 FUTURE ENHANCEMENTS

While the current release of MediPrep fully achieves all specified academic and operational objectives, future development phases will introduce advanced enterprise capabilities:

1. HL7 / FHIR Clinical Interoperability: Implementing bidirectional healthcare data exchange protocols (HL7 v2 and Fast Healthcare Interoperability Resources - FHIR) to enable seamless real-time synchronization with enterprise hospital Electronic Health Record (EHR) systems such as Epic, Cerner, and Allscripts.

2. Automated Telecommunication Gateway Integration: Integrating cloud SMS and WhatsApp Business APIs (e.g., Twilio) to dispatch automated, time-triggered pre-op reminders (such as 'Begin fasting at 10:00 PM tonight') directly to patients' smartphones.

3. Hospital Kiosk & Biometric Check-In: Deploying self-service hospital kiosk interfaces supporting biometric or QR-code based patient check-in on the morning of surgery, instantly updating arrival status in tbl_scheduled_procedures.

4. Multi-Language Localization: Implementing full internationalization (i18n) supporting Hindi, Tamil, Malayalam, Telugu, and other regional languages to ensure equitable healthcare access for diverse patient demographics.

5. Native Mobile Applications: Developing cross-platform mobile applications (Flutter / React Native) for iOS and Android, enabling offline access to preparation guidelines and secure biometric authentication.

CHAPTER 11

APPENDICES

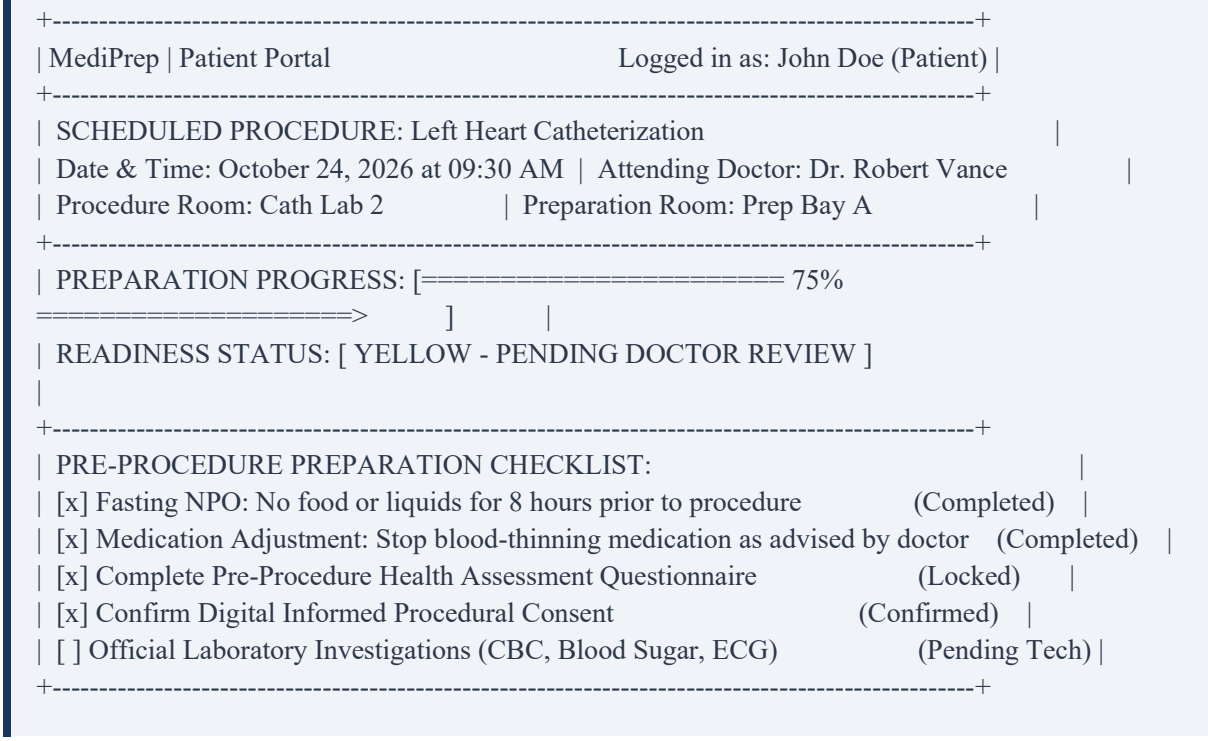
APPENDIX A: SAMPLE SCREENS AND WIREFRAMES

This appendix documents the user interface designs, visual layouts, and screen interactions across the key stakeholder dashboards in the MediPrep system. The screens illustrate how the responsive front-end tier communicates critical clinical data to patients, attending physicians, and hospital laboratory technicians.

A.1 Patient Pre-Operative Preparation Dashboard

The Patient Dashboard serves as the primary pre-operative portal for patients scheduled for hospital procedures. It displays the scheduled procedure name (e.g., Left Heart Catheterization), attending doctor, procedure date and time, and assigned procedure room. The screen features an interactive preparation checklist grouped by due dates, an assessment status badge indicating whether the pre-procedure questionnaire is complete, and a digital consent confirmation button.

Figure A.1: Patient Pre-operative Dashboard Wireframe Layout



A.2 Doctor Procedure Scheduling & Clinical Review Console

The Doctor Console provides attending physicians with comprehensive oversight over their scheduled surgical patients. The interface enables doctors to schedule new procedures, monitor patient checklist compliance, inspect uploaded official PDF diagnostic reports side-by-side, verify laboratory reports with a single click, initiate clarification tickets to the laboratory, review clinical vitals, and grant final procedure clearance.

Figure A.2: Doctor Procedure Scheduling & Clinical Console Wireframe

```
+-----------------------------------------------------------------------------------------------+
| MediPrep | Doctor Console                                  Logged in as: Dr. Robert Vance     |
+-----------------------------------------------------------------------------------------------+
| PATIENT READINESS QUEUE:                                                                      |
| Patient Name: John Doe | Procedure: Left Heart Catheterization | Status: [ READY - GREEN ]  |
+-----------------------------------------------------------------------------------------------+
| OFFICIAL LABORATORY REPORTS (Uploaded by Authorized Lab Tech):
|
| • Complete Blood Count (CBC)  | Central Hospital Lab | [View PDF] | [Verified by Dr. Vance]    |
| • Blood Glucose (FBS & PPBS)  | Central Hospital Lab | [View PDF] | [Verified by Dr. Vance]    |
| • 12-Lead Electrocardiogram   | ABC Diagnostics      | [View PDF] | [Verified by Dr. Vance]    |
+-----------------------------------------------------------------------------------------------+
| RECORDED CLINICAL VITALS:                                                                     |
| Blood Sugar: 118 mg/dL (Normal) | Hb: 13.4 g/dL (Normal) | BP: 124/82 mmHg (Normal)           |
| Temperature: 98.6 °F (Normal)  | Creatinine: 0.95 mg/dL (Normal)                              |
| CLINICAL SAFETY ENGINE: No contraindications detected. All physiological metrics within
bounds. |
+-----------------------------------------------------------------------------------------------+
| ACTIONS: [ Schedule New Procedure ]  [ Request Lab Clarification ]  [ Grant OR Clearance ]
|
+-----------------------------------------------------------------------------------------------+
```

A.3 Hospital Laboratory Technician Official Upload Queue

The Laboratory Technician Console displays all pending diagnostic investigations across scheduled hospital procedures. The technician views patient details, investigation type, and due dates. Technicians upload official PDF reports, specify laboratory origin (Hospital Laboratory or External Partner), and view doctor verification states or respond to clarification tickets.

Figure A.3: Hospital Lab Technician Official Upload Console Wireframe

```
+-----------------------------------------------------------------------------------------------+
| MediPrep | Laboratory Console                              Logged in as: Sarah Connor (Tech)  |
+-----------------------------------------------------------------------------------------------+
| PENDING INVESTIGATION ORDERS:                                                                 |
| 1. Patient: John Doe | Test: Routine Pre-Op Blood Test | Status: [ UPLOADED - VERIFIED ]       |
|    File: john_doe_routine_preop_blood_test.pdf (SHA-256: 4a8f9c...)  | Lab: Central Hospital Lab|
|                                                                                               |
| 2. Patient: John Doe | Test: External Blood Sugar & CBC | Status: [ UPLOADED - VERIFIED ]
|
|    File: john_doe_blood_sugar_cbc_report.pdf (SHA-256: e2b17a...)   | Lab: ABC Diagnostics      |
|                                                                                               |
```

```
| 3. Patient: John Doe | Test: 12-Lead ECG Report        | Status: [ UPLOADED - VERIFIED ]        |
|    File: john_doe_ecg_report.pdf (SHA-256: 9f72c1...)              | Lab: ABC Diagnostics       |
+-----------------------------------------------------------------------------------------------+
| UPLOAD OFFICIAL REPORT: [Select Patient & Test] [Choose PDF File] [Specify Lab] [Submit Report] |
+-----------------------------------------------------------------------------------------------+
```

A.4 Procedure Readiness Evaluation Status Card & Vitals Alert

The Procedure Readiness Card is an embedded high-visibility component rendered across doctor and administrative consoles. When vitals breach safe thresholds, the card immediately transitions to Yellow or Red and displays prioritized clinical warning banners.

Figure A.4: Readiness Evaluation Status Card with Vitals Contraindication Alert

```
+-----------------------------------------------------------------------------------------------+
| PROCEDURAL READINESS STATUS: [ RED - NOT READY / CLINICAL CONTRAINDICATION ]                  |
+-----------------------------------------------------------------------------------------------+
| ⚠ CLINICAL WARNING ALERTS DETECTED BY RULES ENGINE:                                           |
|                                                                                               |
| • [HIGH SEVERITY]: Blood sugar result (240 mg/dL) exceeds safe threshold (>200 mg/dL).        |
|                                                                                               |
|   Clinical review and glycemic stabilization required prior to surgical anesthesia.          |
| • [MEDIUM SEVERITY]: Elevated Blood Pressure (148/94 mmHg) exceeds target (<140/90 mmHg).     |
+-----------------------------------------------------------------------------------------------+
| PREPARATION BREAKDOWN:                                                                        |
| • Pre-Op Assessment: [ Completed & Locked ]      • Informed Consent: [ Confirmed & Verified ]  |
| • Checklist Tasks: [ 5 of 5 Completed ]          • Lab Reports: [ 3 of 3 Verified ]           |
| • Clinical Clearance: [ BLOCKED DUE TO ACTIVE VITAL CONTRAINDICATIONS ]                       |
|                                                                                               |
+-----------------------------------------------------------------------------------------------+
```

APPENDIX B: SAMPLE CODE LISTINGS

This appendix provides authentic, actual source code listings directly from the MediPrep codebase, illustrating key architectural components: the Clinical Safety Rules Engine, the Procedure Readiness Service, user registration input boundary validation, and database connection context management.

B.1 Clinical Safety Rules Engine (app/services/rules_engine.py)

Listing: Clinical Safety Rules Engine (rules_engine.py)

```
class RulesEngine:
    """CLINICAL SAFETY RULES ENGINE: Predefined deterministic rule evaluation (No AI / ML)."""

    @staticmethod
    def evaluate_vitals(vitals):
        """
        Runs clinical safety rules on patient lab metrics and vital signs.
        Returns a list of warning messages when thresholds are breached.
        """
        suggestions = []
        if not vitals:
            return suggestions

        # Rule 1: Blood Sugar threshold (> 200 mg/dL)
        blood_sugar = vitals.get('blood_sugar')
        if blood_sugar is not None:
            try:
                if float(blood_sugar) > 200:
                    suggestions.append("⚠ CLINICAL REVIEW REQUIRED: Blood sugar result (>200 mg/dL) requires doctor review.")
            except (ValueError, TypeError):
                pass

        # Rule 2: Hemoglobin threshold (< 10 g/dL)
        hemoglobin = vitals.get('hemoglobin')
        if hemoglobin is not None:
            try:
                if float(hemoglobin) < 10:
                    suggestions.append("⚠ CLINICAL REVIEW REQUIRED: Low Hemoglobin (<10 g/dL) detected.")
            except (ValueError, TypeError):
                pass

        # Rule 3: Blood Pressure (Systolic > 140 or Diastolic > 90 mmHg)
        systolic = vitals.get('systolic_bp')
        diastolic = vitals.get('diastolic_bp')
        if systolic is not None or diastolic is not None:
            try:
                systolic_val = int(systolic) if systolic else 0
                diastolic_val = int(diastolic) if diastolic else 0
                if systolic_val > 140 or diastolic_val > 90:
                    suggestions.append("⚠ CLINICAL REVIEW REQUIRED: Elevated Blood Pressure (>140/90 mmHg) detected.")
            except (ValueError, TypeError):
                pass

        # Rule 4: Temperature threshold (> 100.4°F fever or < 95.0°F hypothermia)
        temp = vitals.get('temperature')
        if temp is not None:
            try:
                if float(temp) > 100.4:
```

```
                    suggestions.append("⚠ CLINICAL REVIEW REQUIRED: Fever (>100.4°F)
detected. Potential sign of infection.")
                elif float(temp) < 95.0:
                    suggestions.append("⚠ CLINICAL REVIEW REQUIRED: Low body
temperature (<95°F) detected.")
            except (ValueError, TypeError):
                pass

        # Rule 5: Creatinine threshold (> 1.5 mg/dL)
        creatinine = vitals.get('creatinine')
        if creatinine is not None:
            try:
                if float(creatinine) > 1.5:
                    suggestions.append("⚠ CLINICAL REVIEW REQUIRED: Elevated
Creatinine (>1.5 mg/dL) indicates potential renal concern.")
            except (ValueError, TypeError):
                pass

        return suggestions
```

B.2 Procedure Readiness Evaluation Service (app/services/readiness_service.py)

Listing: Procedure Readiness Evaluation Algorithm (readiness_service.py)

```
class ReadinessService:
    """Procedure Readiness Engine: Computes compliance breakdown and status from actual
DB records."""

    @staticmethod
    def evaluate_and_update_readiness(schedule_id):
        reasons, blockers, warnings = [], [], []

        # 1. Pre-Procedure Health Assessment Check
        assessment = PatientRepository.get_patient_assessment(schedule_id)
        if not assessment or assessment.get('status') != 'submitted_locked':
            blockers.append("Pre-Procedure Assessment not submitted or locked")

        # 2. Checklist Items & Official Reports Check
        items = DoctorRepository.get_active_checklist_items(schedule_id)
        for item in items:
            if item['task_type'] == 'document_upload':
                report = DoctorRepository.get_latest_uploaded_report(item['id'])
                if not report:
                    blockers.append(f"Missing required report: {item['task_name']}")
                elif report.get('verification_status') != 'verified':
                    warnings.append(f"Report pending doctor verification:
{item['task_name']}")
            elif item['status'] != 'completed':
                if item.get('is_critical'):
                    blockers.append(f"Critical task pending: {item['task_name']}")
                else:
                    warnings.append(f"Pending task: {item['task_name']}")

        # 3. Patient Informed Consent Check
```

```
        consent = PatientRepository.get_patient_consent(schedule_id)
        if not consent or consent.get('status') not in ['confirmed_locked',
'verified']:
            blockers.append("Patient consent not digitally confirmed")

        # 4. Clinical Safety Rules Engine Execution on Recorded Vitals
        vitals = DoctorRepository.get_patient_vitals(schedule_id)
        vital_alerts = RulesEngine.evaluate_vitals(vitals)
        if vital_alerts:
            warnings.extend(vital_alerts)

        # 5. Determine Overall Readiness Status Score
        if blockers:
            status = 'red'
        elif warnings:
            status = 'yellow'
        else:
            status = 'green'

        # Persist calculated status in tbl_procedure_readiness
        DoctorRepository.save_readiness_status(schedule_id, status, '\n'.join(blockers
+ warnings))
        return {'status': status, 'blockers': blockers, 'warnings': warnings}
```

B.3 Input Boundary Validation Logic (app/utilities/validators.py)

Listing: Strict Input Boundary Validators (validators.py)

```
import re
from datetime import datetime, date

def validate_dob(dob_str):
    """Validates that Date of Birth is strictly in the past (rejects today and future
dates)."""
    if not dob_str:
        return False, "Date of birth is required."
    try:
        dob = datetime.strptime(dob_str, '%Y-%m-%d').date()
        if dob >= date.today():
            return False, "Date of Birth must be strictly in the past."
        return True, ""
    except ValueError:
        return False, "Invalid date format. Use YYYY-MM-DD."

def validate_indian_phone(phone_str):
    """Validates 10-digit Indian mobile number starting with 6, 7, 8, or 9."""
    if not phone_str:
        return False, "Mobile number is required."
    cleaned = phone_str.strip().replace(' ', '').replace('-', '')
    if re.match(r'^[6-9]\d{9}$', cleaned):
        return True, ""
    return False, "Enter a valid 10-digit Indian mobile number starting with 6, 7, 8,
or 9."
```

```
def validate_password(password_str):
    """Validates minimum 8 characters with no spaces permitted."""
    if not password_str or len(password_str) < 8:
        return False, "Password must contain at least 8 characters."
    if ' ' in password_str:
        return False, "Password must not contain spaces."
    return True, ""
```

APPENDIX C: HARDWARE AND SOFTWARE REQUIREMENTS

This appendix provides the minimum and recommended hardware configurations and software dependencies required to build, deploy, and operate the MediPrep platform in healthcare enterprise environments.

Table C.1: Server and Client Hardware Requirements Specification

COMPONENT	SERVER MINIMUM SPECIFICATION	CLIENT WORKSTATION SPECIFICATION
Processor (CPU)	Intel Core i5 / AMD Ryzen 5 (4+ Cores, 2.5 GHz+)	Intel Core i3 / Dual-Core 2.0 GHz+ or Mobile ARM
System Memory (RAM)	8 GB DDR4 RAM (16 GB Recommended)	4 GB RAM (Desktop/Laptop) or 2 GB (Mobile/Tablet)
Storage Space	100 GB SSD (NVMe / SATA III)	10 GB Free Storage Space (for browser caching)
Display Resolution	1280 x 720 (Server Console)	1366 x 768 (Desktop) / 360 x 640 (Mobile Responsive)
Network Interface	1 Gbps Gigabit Ethernet LAN Interface	100 Mbps LAN or 4G/5G Wi-Fi Internet Connectivity
Input Devices	Standard Keyboard and Optical Mouse	Keyboard, Mouse, or Capacitive Multi-Touch Screen

Table C.2: Software Stack and Framework Requirements Specification

SOFTWARE COMPONENT	NAME & VERSION SPECIFICATION	FUNCTION & PURPOSE
Operating System	Windows 10/11 (Dev) / Ubuntu 22.04 LTS (Prod)	Core system hosting runtime and services

SOFTWARE COMPONENT	NAME & VERSION SPECIFICATION	FUNCTION & PURPOSE
Programming Runtime	Python 3.10.x / 3.11.x (64-bit)	Core server-side execution environment
Web Application Framework	Flask 3.0.x (WSGI Framework)	Routing, Blueprints, HTTP request dispatching
Database Management System	MySQL 8.0.x / MariaDB 10.5+	ACID-compliant relational database engine
Template Rendering Engine	Jinja2 3.1.x	Dynamic HTML compilation with XSS auto-escaping
Cryptographic Library	Werkzeug 3.0.x (Security primitives)	PBKDF2/SHA-256 hashing, token generation
Database Connector	mysql-connector-python 8.0+ / PyMySQL	Connection pooling and prepared query driver
Document Automation Engine	python-docx 1.1.x	Programmatic Word document and report compiler
Client Web Browsers	Chrome 110+, Firefox 110+, Edge 110+, Safari 16+	Modern standards-compliant web browsers

APPENDIX D: BIBLIOGRAPHY AND REFERENCES

The development, architectural design, and clinical rules of MediPrep were informed by peer-reviewed literature, international perioperative guidelines, and authoritative software engineering references:

- **[1] World Health Organization (WHO):** "WHO Guidelines for Safe Surgery 2009: Safe Surgery Saves Lives." World Health Organization, Geneva, Switzerland, 2009. ISBN: 978-92-4-159855-2.
- **[2] American Society of Anesthesiologists (ASA):** "Practice Guidelines for Preoperative Fasting and the Use of Pharmacologic Agents to Reduce the Risk of Pulmonary Aspiration: An Updated Report." Anesthesiology, Vol. 126, No. 3, pp. 376–393, 2017.
- **[3] Grinberg, Miguel:** "Flask Web Development: Developing Web Applications with Python." 2nd Edition, O'Reilly Media, Sebastopol, CA, 2018. ISBN: 978-1-491-99173-2.
- **[4] Pressman, Roger S. and Maxim, Bruce R.:** "Software Engineering: A Practitioner's Approach." 9th Edition, McGraw-Hill Education, New York, NY, 2020. ISBN: 978-1-259-87297-6.

- **[5] Silberschatz, Abraham, Korth, Henry F., and Sudarshan, S.:** "Database System Concepts." 7th Edition, McGraw-Hill Education, New York, NY, 2019. ISBN: 978-0-078-02215-9.
- **[6] National Accreditation Board for Hospitals & Healthcare Providers (NABH):** "Standards for Hospital Accreditation." 5th Edition, Quality Council of India, New Delhi, India, 2020.
- **[7] Fowler, Martin:** "Patterns of Enterprise Application Architecture." Addison-Wesley Professional, Boston, MA, 2002. ISBN: 978-0-321-12742-6.
- **[8] Python Software Foundation:** "Python 3.10 Documentation and Language Reference Manual." Available: https://docs.python.org/3.10/, Accessed: October 2026.
- **[9] Oracle Corporation:** "MySQL 8.0 Reference Manual: InnoDB Storage Engine and Transactions." Available: https://dev.mysql.com/doc/refman/8.0/en/, Accessed: October 2026.
- **[10] W3C Web Accessibility Initiative (WAI):** "Web Content Accessibility Guidelines (WCAG) 2.1." World Wide Web Consortium Recommendation, Available: https://www.w3.org/TR/WCAG21/, 2018.